Lateral analysis – Level 2 – Damped natural frequency analysis – OH and BB pumps with more than two stages

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1. Purpose.

This document is related to the dynamic analysis requirements for rotordynamic pump, especially to lateral analysis.

The procedure to define what level of detail for the dynamic analysis is required to be in accordance with good practice of pump design is described in *G.10.00.09* and *ANSI/HI 9.6.8-2014*: if calculated RUN is higher than 20 and less equal to 50, a **Level 2 – Intermediate method – Lateral rotordynamic analysis – Damped natural frequency analysis** is required**.**

This document describes how to perform a damped natural frequency (lateral) analysis.

2. Scope.

This procedure is applicable to OH or BB pumps with more than two stages regardless of bearing type, also one- and two-stage pumps with hydrodynamic bearings.

The procedure is applicable for the Nova Milanese and La Spezia plants.

3. Definitions.

B	bearing width
BB	between bearings (API610)
D, d	diameter
D_2, d_2	impeller diameter
D_B	bearing diameter
D_{sp}, d_{sp}	diameter annular seal
d, d_q	direct and cross coupling damping coefficient
DOF	degree of freedom
dR	radial clearance
dS	pad clearance
F, f	force or frequency
FDC	fluid coefficient
FEA	finite element analysis

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g_a	radial distance between outer impeller and casing
GUI	graphical user interface
H	head per stage
H_P	static pressure rise in impeller
IDmb	inner bending mass diameter
IDsb	inner bending stiffness diameter
k, k_q	direct and cross coupling stiffness coefficient
k	rotation factor
ISR	impeller side room
L	length
L_S	height front shroud
L_{sp}	length annular seal
M, m	mass
m, m_q	direct and cross coupling mass coefficient
m	preload
N	rotating speed
n_q	specific speed
ODmb	outer bending mass diameter (Madyn 2000)
ODsb	outer bending stiffness diameter (Madyn 2000)
OH	overhung (API610)
p	pressure
PLB	product lubricated bearing
q*	flow rate referred to flow rate at best efficiency point
Q_{s1}, Q_{s2}	leakage flow rate at front wear ring
Q_{s3}	leakage flow rate at interstage seal
R	risk number (ANSI HI 9.6.8-2014)

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r_2	impeller radius
R_G	degree of reaction
R_s	pad radius
Re	Reynolds number
REB	rolling Element Bearing
RFB	radial fluid film bearing
RUN	R multiplied by U (ANSI HI 9.6.8-2014)
s_{ax}	axial distance between impeller shrouds and casing
TLA	Torsional Lateral Axial Analysis
U	uncertainty value (ANSI HI 9.6.8-2014)
VS	vertically suspended (API610)
β	angular velocity of fluid
ρ	density
ψ	bearing relative clearance
ψ_v	pad clearance ratio
ω	rotating speed

Campbell diagram it indicates the natural frequencies within the excitation natural frequency range and displays the margin between potential torsional resonances identified the analysis versus those excitation frequencies (1x running speed, 2x running speed, vane pass, 1x and 2x electrical line frequency, etc…).

Critical Speed shaft rotational speed at which the rotor-bearing-support system is in a state of resonance.

Station boundary points of a discrete element.

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4. General procedure: description.

In OH or BB pumps, a lateral rotodynamic analysis is performed to assess the potential for high vibration, associated degradation, and eventually equipment failure after installation. Excessive lateral vibration can lead, or contribute, to the following types of problems:

- Accelerated wear at internal wear ring clearances and loss of performance;
- Rubs at internal locations;
- Accelerated bearing wear;
- Excitation of high vibration associated with system structural modes, leading to possible fatigue of structural components.

Lateral critical speed analysis for OH or BB pumps is performed for the combined pump rotating components (shaft, impellers, sleeves, coupling half, etc.) up to the first flexible shaft coupling (including the half-coupling), or for the entire pump-driver rotor system if a rigid coupling is used. Because of the strong effects of seal damping in most centrifugal pumps, the damped analysis is the most relevant for most of pumping applications, particularly for pumps of more than two stages.

The entire pump rotor should be included in the model. This includes the rotating component dimensions, weights, and inertias, as well as support stiffness from bearings, and internal seal clearances. The shaft cross-sectional inertial and mass properties are calculated from the length, outer diameter, and inner diameter of each shaft section from the shaft drawing. Impellers are modelled as added masses and rotational/transverse inertias at referenced shaft locations. Gyroscopic effects are included by the action of the component polar and transverse mass moment of inertia. This is particularly important with OH pumps because the impeller produces a restoring moment when whirling in a deflected position. A flexible coupling may isolate the lateral vibration between the pump and the driver. As per API RP684 best practice, the coupling half-weight (including the spacer, if any) is modelled at the centre of gravity of the half-coupling. The mathematical representation of shaft elements should include not only bending stiffness, but also transverse shear stiffness.

Hydrodynamic bearing effective support stiffness is determined from lube oil pressure, lubricant characteristics, and bearing geometry/clearance, with the result that the bearings are represented as two-dimensional spring and damping coefficients in the model including the so-called “cross-coupled” stiffness that represents reaction force build up in the direction perpendicular to the shaft motion instead of in-line with it (“direct” stiffness). Stiffness versus load curves provided by the bearing manufacturer are used to input rolling element bearing stiffness. The impact on critical

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speeds, damping and cross-coupling from annular seals (grooved or non-grooved) and pressure-reducing devices used in multistage pumps are accounted for in the analysis Changes in these devices can impact critical speed values and definitely have a positive or negative impact on damping. The structural support stiffness from bearing mounting components, referenced to ground, should be included if the stiffness of these components is less than 10 times the stiffness of the bearing itself.

The analyst calculates the natural frequencies and characteristics of each mode (mode shape, damping, etc.). Fluid flowing/leaking through wear rings (Lomakin effect) or the specifics of a bearing design affects the stiffness and damping of rotor system. If the natural frequency is not well damped (refer to API RP684 or API Annex I) and it is located near a potential excitation frequency (1X, 2X, vane pass, line frequency, etc.), then an iteration of the analysis should be performed such that suitable recommendations can be made to correct the issue. Iteration may involve a shaft diameter, impeller weight, coupling, or bearing geometry modification.

A lateral analysis should be performed at the design point with “as-new” operating clearances, and include an iteration considering “worn” operating clearances. As a point of reference, API 610 requires the worn clearance analysis to be performed at twice the “as-new” clearances. In addition, a specification may call for a lateral forced response analysis. This is performed by taking the completed lateral analysis and applying an expected worse case or specification maximum acceptable imbalance to the rotor to determine the amplitude of any vibration (taking into account any resonances and the predicted damping or a particular mode shape).

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5. Methodology, analysis and interpretation of results.

Methodology.

Software/Tools:

- general purpose FEA programs or specialty rotordynamic programs;

The tool described in this procedure is:

- Madyn 2000

Model.

Mass-elastic model of pump rotor only (low lateral stiffness of flexible disk type couplings decouple pump rotor from motor rotor with regards to lateral rotordynamic behavior).

Shaft segments modeled with beam and pipe elements.

Elements have four nodal degrees of freedom:

- Translational in both lateral directions (horizontal and vertical; axial is constrained at the thrust bearing) and rotation around lateral axes (torsion is constrained at all nodes).

Impeller and pump coupling half (pump coupling hub and half of coupling spacer) are generally considered as lumped mass effects (mass and inertia) without stiffness contribution. Impellers do not significantly contribute to the overall rotor stiffness in cases of a loose fit on the shaft. Impellers mounted with an interference fit may contribute to the overall rotor stiffness, depending on the level of interference. Bracketing calculations may be performed to evaluate the influence of the interface fit on the rotor rigidity.

Shaft sleeves, rotating portion of the mechanical seal, etc. are considered as distributed mass effects or lumped mass in their COG (no stiffness contribution).

Interaction forces from annular seals are considered.

Impeller-casing interaction considered.

Liquid effects from fluids contained within impeller passageways, and fluids displaced by wetted component volume are considered).

Nonrigid radial bearing stiffness and, depending on the design of bearing housing/ bearing hanger/ baseplate, nonrigid bearing support stiffness considered.

Analysis.

First, a dry natural lateral analysis (no liquid effects) should be performed reporting the first, second and third “dry” bending natural frequencies.

Second, damped natural frequency analysis should be performed within 25% to 125% of continuous operating range and rotor’s damped natural frequencies within the frequency range zero to 2.2 times maximum continuous speed should be reported taking account of the following conditions:

- Internal running clearances “as new”, with water;
- Internal running clearances “as new”, with pumped liquid;
- Internal running clearances “twice the as new”, with pumped fluid.

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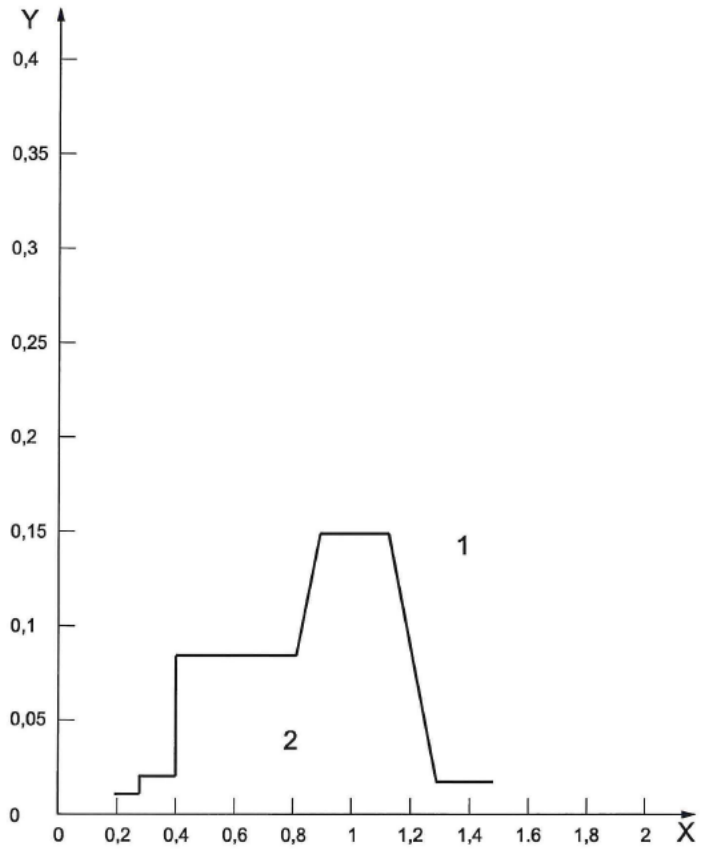
Interpretation of results.

Results of a damped natural frequency analysis should include:

- mode shapes, modal damping and corresponding natural frequency.
- Campbell diagram exhibiting 1X synchronous speed excitation as well as significant modes..
- Damping factor versus separation margin between any bending natural frequency and the synchronous run line (when liquid effects are included).



Key
X frequency ratio, f_{ni}/f_{run}
Y damping factor, ξ
1 acceptable region
2 unacceptable region

Figure 1. Damping factor vs Frequency ratio.

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5.1. Modelling shaft

Figure 2 shows a uniform steel beam 20 in. long, 1 in. in diameter, and with an undamped bearing at each the end having K = 1000 lb/in, *Vance et al*. The mode shapes of the first four modes are also shown. The first two modes are fairly rigid. Modes like these don't require many elements for good accuracy. The second two modes look like a half sine wave and a full sine wave.

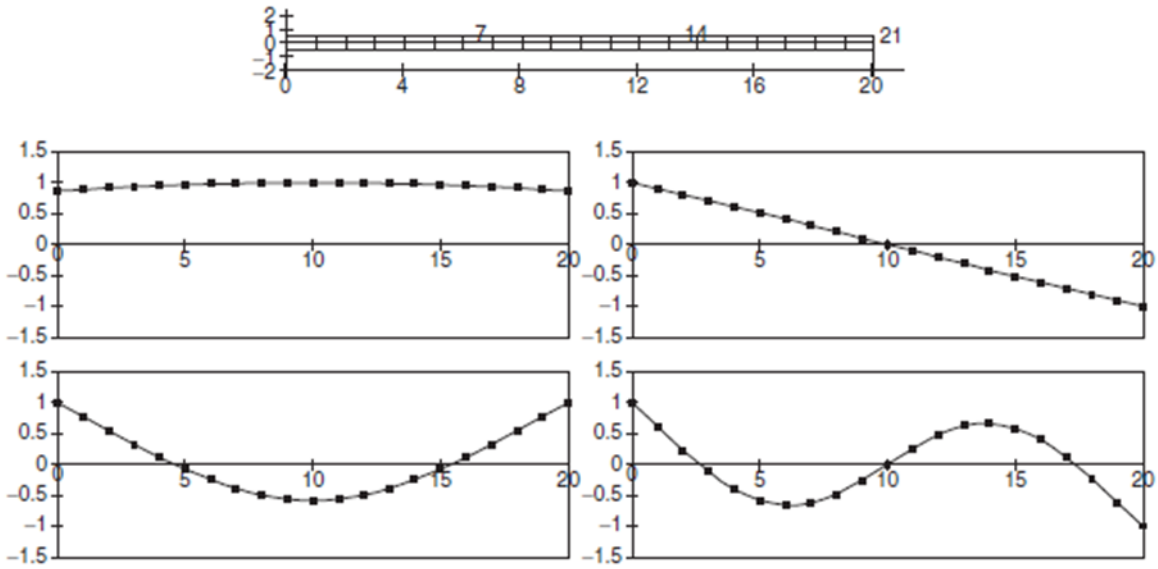

Figure 2 Model used for mode accuracy study, and mode shapes of the first four normal modes.

Figure 3 shows that accuracy depends not only on the number of elements, but also on the formulation of the elements. Notice in Figure 3Figure 3 that the rigid modes have very small percentage errors. This is because there is little to no flexure of the elements. The flexible modes, however, illustrate the effect of the number of elements and also the difference between the four different element types. For our uniform beam test rotor, the four-element case corresponds to an L/D of 5. For this case the first mode, which is a rigid shaft mode, has frequency errors less than 1 percent. Even the second mode, which is also a rigid shaft mode, has errors less than 1 percent for all but the uncorrected 2-node element. For the first flexible mode, however, four elements are not enough, and the errors exceed 10 percent. Based on the errors computed for this example, a guideline of "**use elements with L/D < 1**" will result in acceptable accuracy for all cases considered. Another reason to maintain an element L/D < 1 is that when using a transfer matrix-based calculation procedure, the calculations can become numerically ill-conditioned if the L/D of very many elements is much longer than one. This is an aspect specific to the algorithm, and not the rotor system model.

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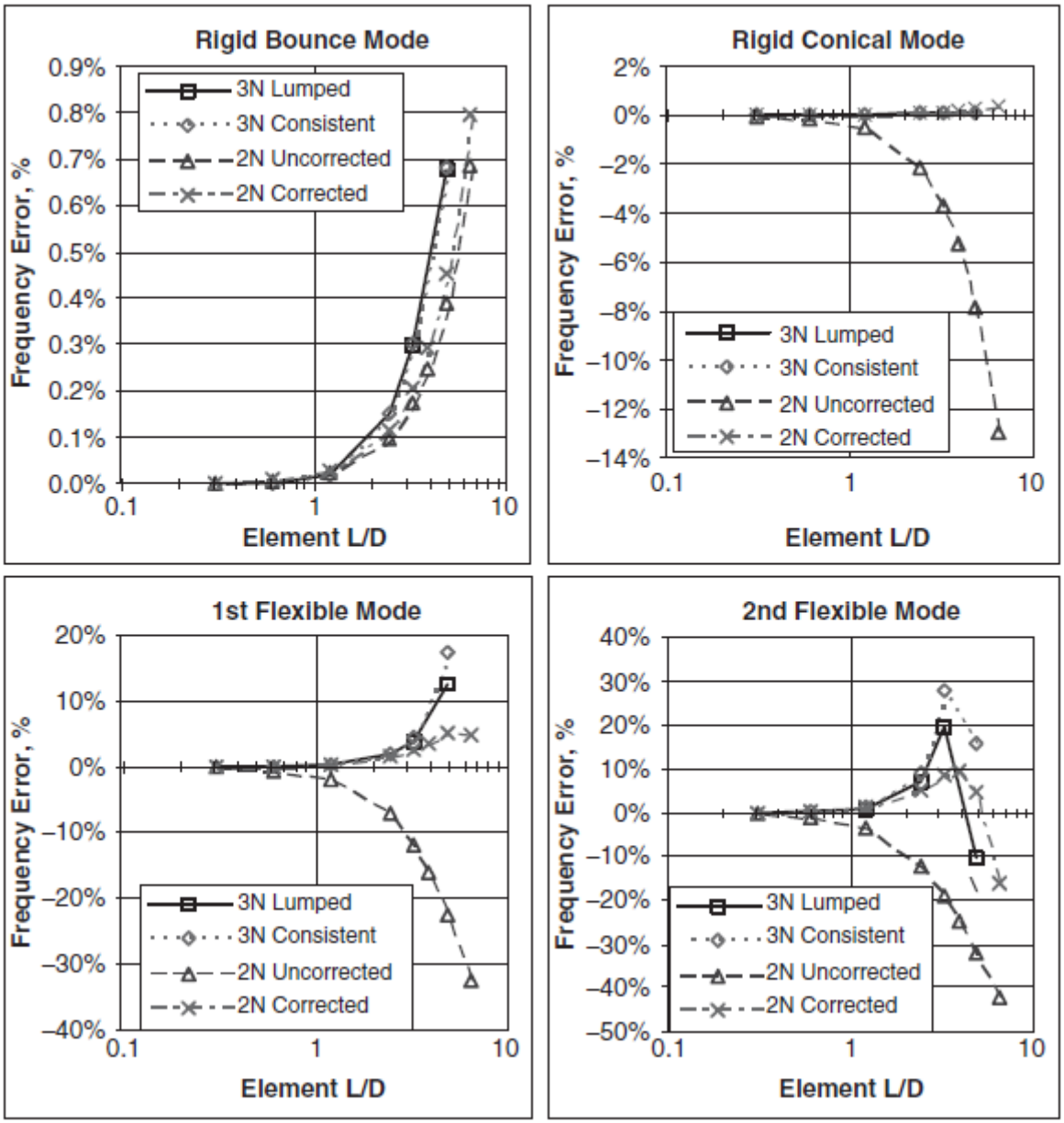

Figure 3 Mode accuracy versus number of elements for four different element definitions.

Another aspect to be aware when modelling the shaft is the **pumping fluid temperature**. Temperature influences mechanical properties that characterize the strength of the material, such as Young's modulus. ASME Section II, Part D gives some reference to moduli of elasticity of different ferrous materials at different temperature.

The recommendation is to assign the proper mechanical properties when performing the analysis.

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5.2. Modelling annular seals

Forces created in the annular seals at the impeller or at an axial thrust balance device have an altogether dominating impact on pump rotor vibrations. To a large extent they determine the eigen frequencies, eigen modes, amplitudes of forced response and the stability limit of multistage pumps. The fluid forces created in an annular seal are usually calculated according to a linearized model. The radial and tangential force components are described by spring (k), damping (d) and mass coefficients (m) which are proportional to the displacement, the velocity and the acceleration of the rotor movement:

$$-\begin{bmatrix} F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} k & k_q \\ -k_q & k \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} d & d_q \\ -d_q & d \end{bmatrix} \begin{bmatrix} \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} + \begin{bmatrix} m & m_q \\ -m_q & m \end{bmatrix} \begin{bmatrix} \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix}$$

- k, direct stiffness coefficient
- k_q, cross coupling stiffness coefficient
- d, direct damping coefficient
- d_q, cross coupling stiffness coefficient
- m, direct mass coefficient
- m_q, cross coupling mass coefficient
- F_2 force in vertical direction
- F_3 force in horizontal direction

The resulting forces can be interpreted as integrals over the pressure distribution and wall shear stresses in the seal. Equation above was derived for oscillations around a central rotor position, but it can be used approximately for static eccentricities up to 50% of the seal clearance. In the equation rotational symmetry was assumed (the diagonal terms in the matrices are equal). This assumption is essentially fulfilled for short seals with $L_{sp}/d_{sp} < 0.5$. For long seals with $L_{sp}/d_{sp} > 0.7$, such as a balance piston, a fourth matrix must be added to account for the tilting of the rotor axis in the seal. The fourth matrix is not needed if the long seal is interrupted by deep annular grooves which allow a pressure equalization around the circumference.

The force in radial direction is determined by the direct stiffness, because it dominates over the mass term and the cross-coupled damping. The direct stiffness of a short annular seal (say $L_{sp}/r_{sp} < 0.3$) is caused by the following mechanism: the flow velocity on the side with the smaller gap (due to the higher flow resistance) is lower than on the side with the larger clearance, Figure 4. Due to the greater dynamic pressure and inlet loss on the side with the higher velocity, a pressure difference between the narrow and the wide gap is created. This pressure difference generates a force which

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is opposed to the rotor displacement and thus has a centering effect on the rotor. The mechanism is frequently referred to as the “Lomakin effect”.

The centering “Lomakin effect” is opposed by a “Bernoulli effect”, Figure 4. For reasons of continuity the circumferential flow, induced by shear stresses on the rotating shaft, has its velocity maximum on the side with the narrow gap. According to the Bernoulli equation the highest velocity corresponds to the lowest pressure. As a consequence, a de-centering force component is acting from the wider to the narrower gap. Since it takes some distance to generate an appreciable circumferential flow by wall shear stresses, the Bernoulli effect is weak in short seals with low pre-swirl, but it becomes dominating in long annular seals. The counter-acting Lomakin and Bernoulli effects lead to a maximum of the direct stiffness which can be expected around L/D = 0.5. The foregoing statement is true only if the fluid entering the seal does not have a strong circumferential velocity. Very long seals which are not interrupted by deep annular grooves are de-centering.

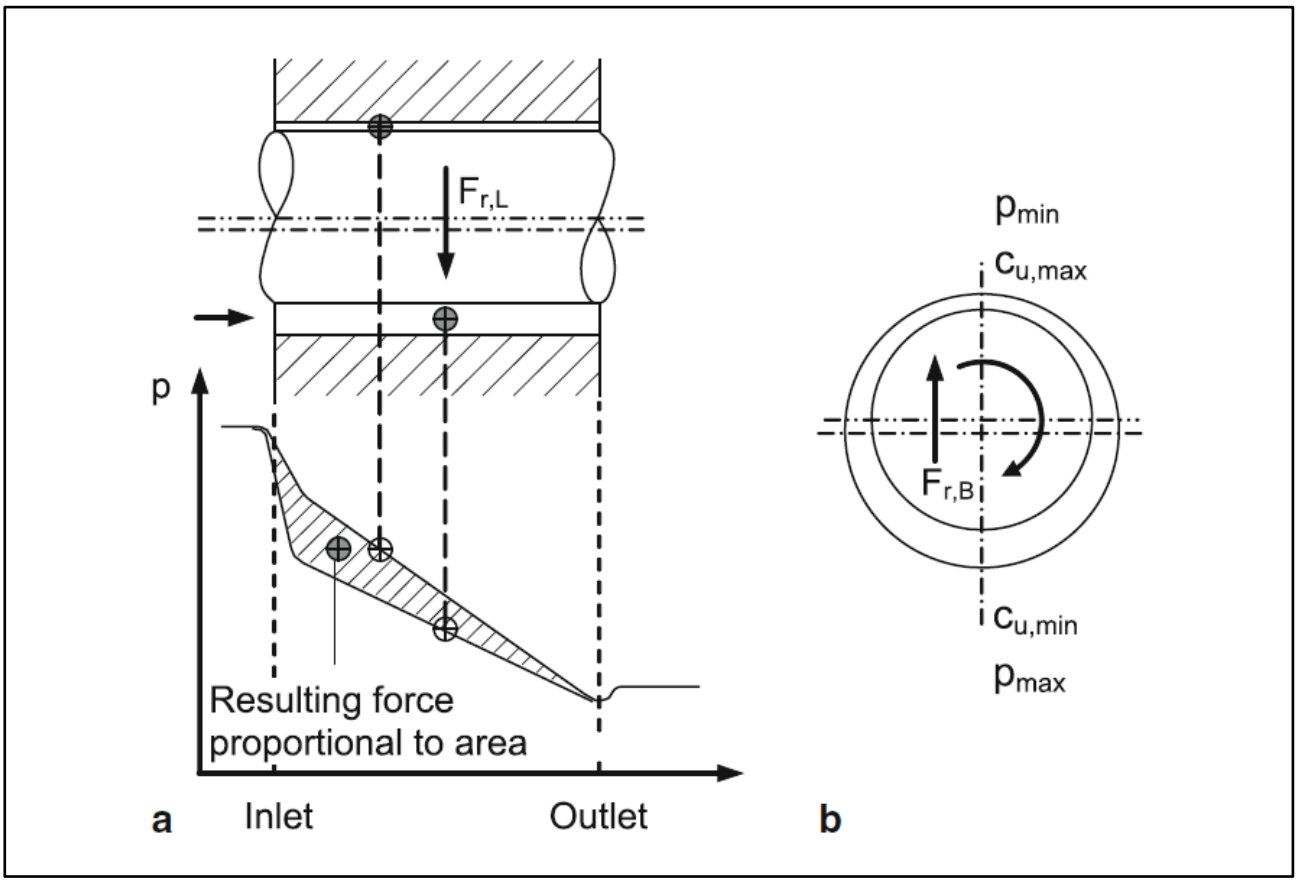

*Figure 4. **a** Effect of axial flow (Lomakin effect). **b** Effect of circumferential flow (Bernoulli effect).*

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5.2.1. Pressure drop across the seal calculation

Coefficients are proportional to the pressure difference across the seal. It can be established starting from the static pressure H_P prevailing at the impeller outlet. In many cases H_P can be estimated accurately enough with the help of the degree of reaction from $H_P = R_G \cdot H$. For moderate specific speed (n_q < 40) R_G = 0.75 is a good assumption. With radially inward direct leakages (Q_{s1} and Q_{s2}) the pressure between the impeller outlet and the seal drops due to the rotation of the fluid in the impeller sidewall gap. This is described by the rotation factor $k = \beta/\omega$ (β is the angular velocity of the fluid).

The pressure reduction in the impeller sidewall gap can be calculated from $\frac{1}{2}\rho k^2 u_2^2 \cdot \left(1 - \frac{{d_{sp}}^2}{d_2^2}\right)$; then, the pressure difference over the seal at the impeller inlet results:

$$\Delta H_{sp} = \left(H_P - k^2 \frac{u_2^2}{2g} \cdot \left(1 - \frac{{d_{sp}}^2}{d_2^2}\right)\right)$$

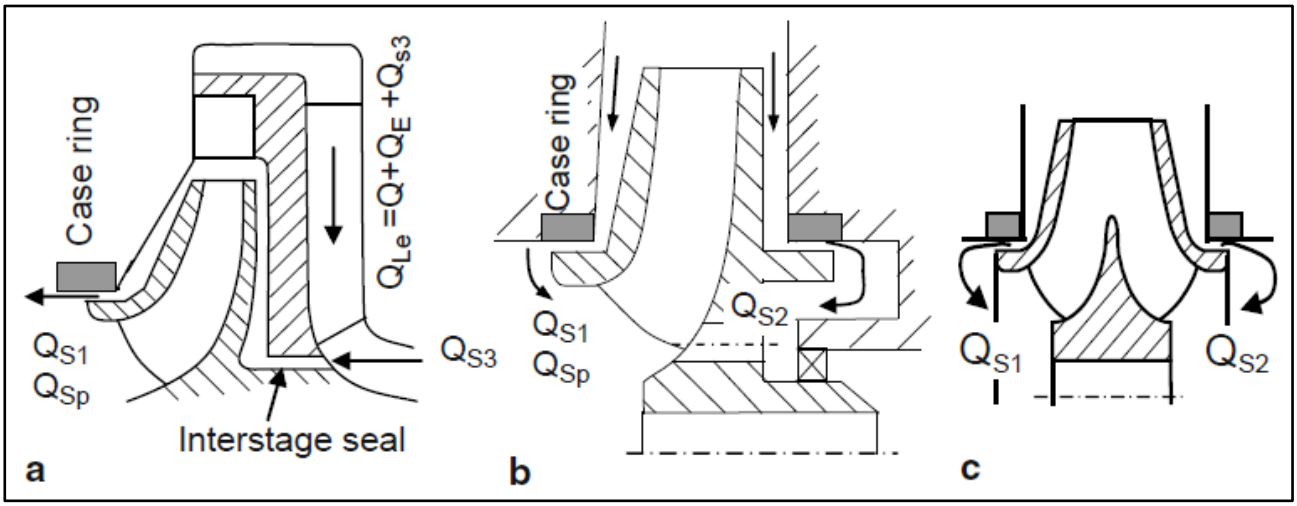

Figure 5. Leakage flows: a) multistage pump; b) impeller with balance holes; c) double entry impeller.

If the leakage Q_{S3} flows radially outward as in the case of the interstage seals of multistage pumps, the pressure differential in the sidewall gap is added to the pressure recovery in the diffuser:

$$\Delta H_{S3} = H - \left(H_P - k^2 \frac{u_2^2}{2g} \cdot \left(1 - \frac{{d_{sp}}^2}{d_2^2}\right)\right)$$

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The rotation factor is estimated as a function of the Reynolds number and the seal geometry as shown in .

Table 1. Rotation factor k for leakage radially inward and outward.

Q_{sp} radially inward	$k = 0.9 y_{sp}^{0.087}$	$y_{sp} = Re_{u_2}^{0.3} \cdot \frac{s \cdot d_{sp}}{{d_2}^2} \cdot \sqrt{\frac{s}{L_{sp}}}$
Q_{sp} radially outward	$k = 0.24 y_{sp}^{-0.096}$	$Re_{u_2} = \frac{u_2 \cdot d_2}{2\nu}$

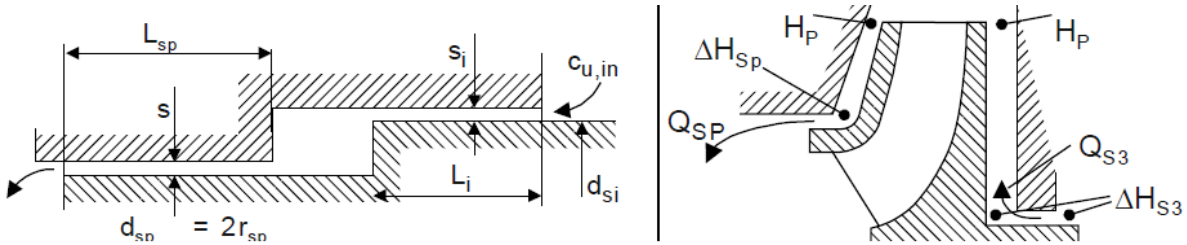

Figure 6. Seal geometry, pressure drop and leakage.

In this way, the pressure drop across the seal can be estimated.

Refer to Mod.493 – Pressure Drop across the Seal to estimate the inlet and outlet pressure to assign. The pressure drop will be scaled to adapt at the different rotating speed required by the lateral analysis (25% - 125% rotating speed).

$$\Delta p = \Delta p_{rated} \cdot \left(\frac{\omega}{\omega_{rated}}\right)^2$$

5.2.2. Relative inlet swirl

Inlet swirl velocity has an important impact on seal coefficients. Although this parameter depends on the specifics of the configuration, Florjancic and McCloskey state that the swirl ratio at the entrance of most wear rings is about 0.70 to 0.80. Additionally, Childs states that most interstage seals that are downstream of diffuser have inlet swirls that are negligible.

Inlet swirl can be reduced or eliminated via placement of a swirl brake at the inlet. Two alternative configurations of swirl brakes are available radial slots or radial bores. Radial slot swirl brakes are a conventional type consisting of a certain number of radial slots placed directly in front of balance piston (inline design) or center bushing (back-to-back design). The second type consists of radial bores machine on the static part of the bushing, which injects fluid in the radial direction.

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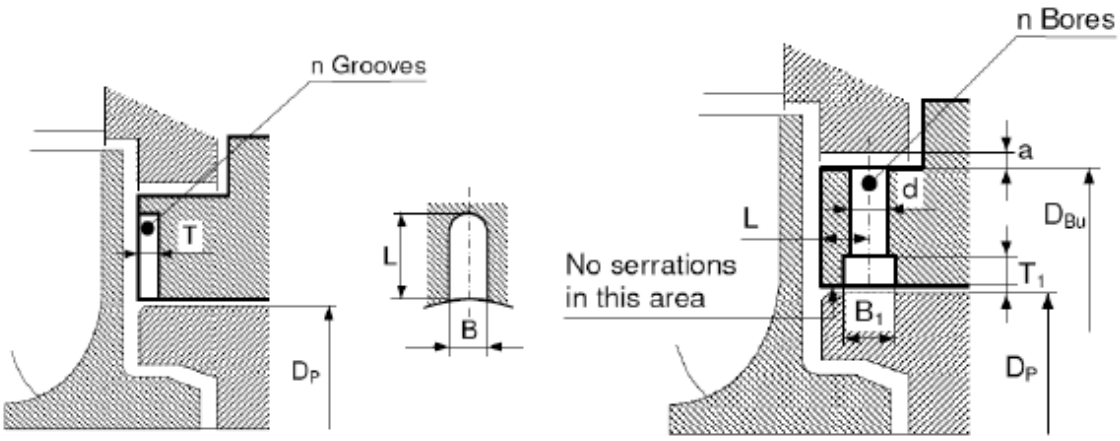

Figure 7. Swirl brakes.

In case of swirl brakes, put the inlet swirl to 0%.

With long seals (L_{sp}/D_{sp} > 0.7), tilting of the rotor versus the bushing must be considered because it impairs the stiffness of the seal. Therefore, deep annular grooves are machined in the balance piston bushing at two or three locations along its length in order to foster pressure equalization around the circumference of the piston. The force coefficients can be estimated modelling the entire geometry or separately for the 2/3 section. In the latter case the inlet swirl to use is 50% starting from the second section. In Figure 8, section 1 has 50% inlet swirl because it is related to the values in Table 2. In case of separated sections, it is not required to follow the procedure described in "*5.2.3 Long seals*" since L_{sp}/D_{sp} will be less than 0.7.

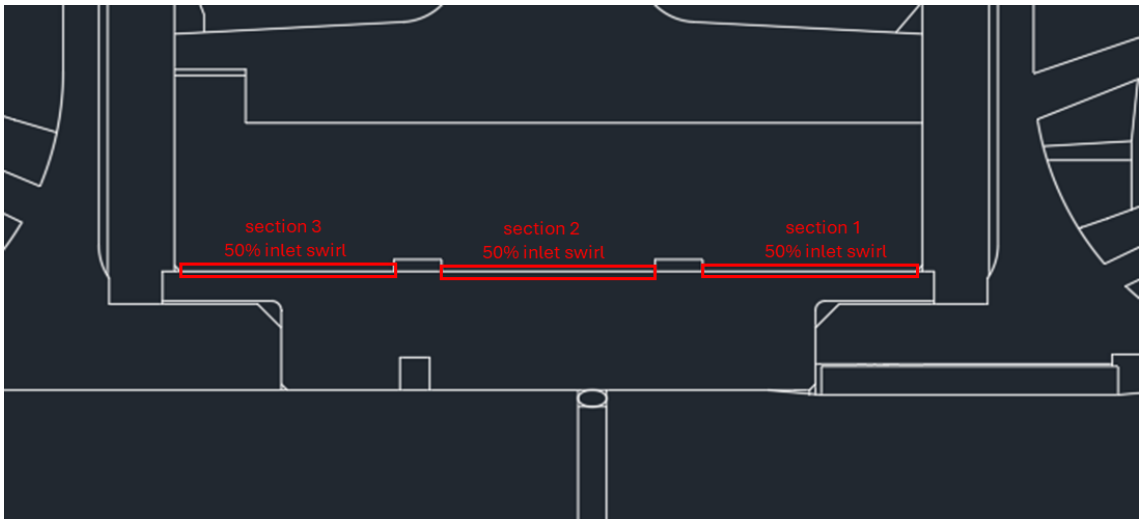

Figure 8. Modelling long seal in separated sections.

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Table 2 summarizes the inlet swirl for the different seals.

Table 2. Inlet swirl seals.

	Flow	Inlet swirl [%]
Wear Ring	Radially inward	70 (or 80)
Interstage Seal	Radially outward	0
Balance piston	Radially inward (inline pump)	50
	- (back-to back pump)	0
Central bush	Radially inward	50
Swirl Brake	-	0
Inlet Deep Groove	-	50

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5.2.3. Long seals

For long seals, smooth or with circumferential grooves, with L_{sp}/D_{sp} > 0.7, such as a balance piston, fourth rank matrixes must be used in order to account for the tilting of the rotor axis in the seal. The fourth matrix is not needed if the long seal is interrupted by deep annular grooves which allow a pressure equalization around the circumference and the geometry is spit in 2/3 separated sections as shown in Figure 8.

In software that do not consider rotating degree of freedom, such as Madyn 2000, include the tilting effect of the rotor, follow this procedure:

- Estimate dynamic coefficients of the full-length annular seal (L);
- Halve the resulting coefficients;
- Define two stations at distance $L^* = \frac{L}{2\sqrt{3}}$ from annular seal center (Figure 9) and apply at these stations the halved coefficients.

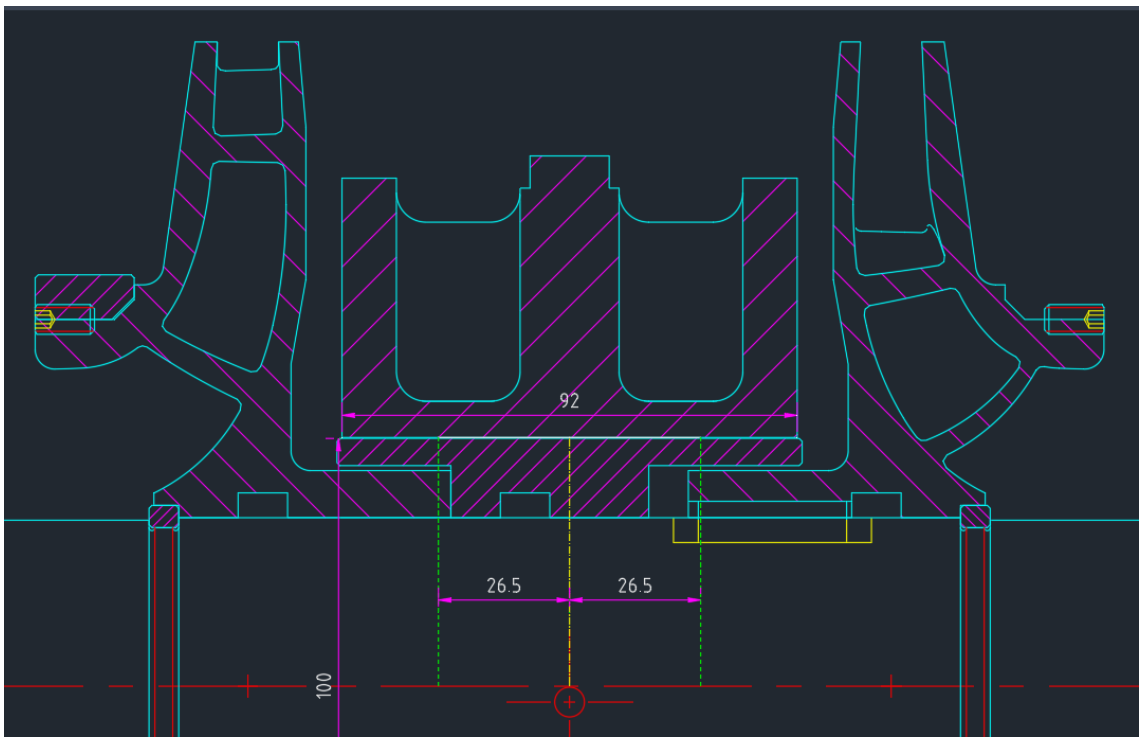

Figure 9. Example of splitting a long seal.

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5.3. Hydraulic Impeller Interaction

If an impeller moves around its centric position due to vibrations of the rotor, the pressure distribution over the impeller circumference changes. Reaction forces are thus generated in a similar way as with annular seals. The interaction forces do not only act on the impeller outlet width but also on the shroud. If the shroud is not perpendicular to the rotor axis, this effect can be considerable.

The radial force resulting has little impact on the eigen frequencies because the direct stiffness is low and the effects of the mass and cross-coupled damping (both are significant) compensate to a large extent. Overall, the effect of the hydraulic impeller interaction on the eigen frequencies is slightly negative (centering). The tangential force is destabilizing because the cross-coupled stiffness is high while the direct damping is weak. Typically, the impeller hydraulic interaction is one order of magnitude smaller than the effect of the annular seals.

Measuring a given impeller once in combination with a volute and then with a diffuser revealed little differences in the impeller interaction coefficients. Likewise, differences between vaned and vaneless diffusers were insignificant. The hydraulic impeller interaction is mainly influenced by the front shroud. To minimize its destabilizing effect, the radial projection of the impeller front shroud should expose an area to radial forces as small as possible.

The coefficients grow with decreasing sidewall clearance. The cross-coupled stiffness, and thus the destabilizing effect of the front shroud, increases with growing pre-rotation. The leakage flowing through the impeller sidewall gap affects primarily the direct stiffness, which has a de-centering effect and thus reduces the eigen frequency.

Based on the test data published, correlations (Table 3) have been developed which allow to estimate the impeller interaction forces as function of geometry, Reynolds number and flow rate ratio q*.

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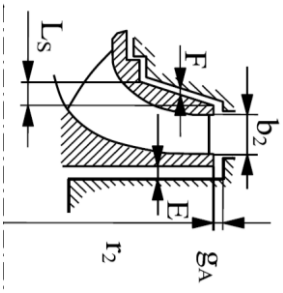
For high-head centrifugal pumps, Bolleter tests appear to be the most relevant in terms of geometry, Reynolds number and inlet swirl to the impeller side room (ISR). The correlations cover the range of:

- $Re = 7 \times 10^5 \; to \; 7 \times 10^7$
- $q^* = 0.25 \; to \; 1.4$
- $L_s/r_2 = 0 \; to \; 0.25$
- $s_{ax}/r_2 = 0.016 \; to \; 0.18 \; (strongly\ non-linear, s_{ax} = F)$
- $g_a/r_2 = 0.0045 \; to \; 0.06 \; (strongly\ non-linear)$
- $d_2 = 160 \; to \; 370 \; mm$

Table 3. Hydraulic impeller interaction dimensionless coefficients.

Direct stiffness	$k^* = \frac{-3.4}{0.53 + 0.47q^*} \cdot (0.25 + 6L_s{}^*) \cdot \left(\frac{0.033}{s_{ax}{}^*}\right)^{0.32} \cdot \left(\frac{0.02}{g_a{}^*}\right)^{0.55}$	
Cross-coupled stiffness	$k_c{}^* = 4.1 \cdot \left(\frac{10^7}{Re}\right)^{-0.25} \cdot (0.67 + 2.64L_s{}^*) \cdot \left(\frac{0.033}{s_{ax}{}^*}\right)^{0.4} \cdot \left(\frac{0.02}{g_a{}^*}\right)^{0.2}$	
Direct damping	$c^* = 3.8 \cdot \left(\frac{10^7}{Re}\right)^{0.1} \cdot (0.67 + 2.64L_s{}^*) \cdot \left(\frac{0.033}{s_{ax}{}^*}\right)^{0.5} \cdot \left(\frac{0.02}{g_a{}^*}\right)^{0.55} \cdot (1.67 - 0.67q^*)$	
Cross-coupled damping	$c_c{}^* = 10.4 \cdot \left(\frac{10^7}{Re}\right)^{0.2} \cdot (0.67 + 2.64L_s{}^*) \cdot \left(\frac{0.033}{s_{ax}{}^*}\right)^{0.7} \cdot \left(\frac{0.02}{g_a{}^*}\right)^{0.25} \cdot (1.67 - 0.67q^*)$	
Direct mass	$m^* = 8.3 \cdot \left(\frac{10^7}{Re}\right)^{0.1} \cdot (0.25 + 6L_s{}^*) \cdot \left(\frac{0.033}{s_{ax}{}^*}\right)^{0.65} \cdot \left(\frac{0.02}{g_a{}^*}\right)^{0.17} \cdot (1.33 - 0.33q^*)$	
Cross-coupled mass	$m_c{}^* = 3 \cdot \left(\frac{10^7}{Re}\right)^{-0.55} \cdot (0.25 + 6L_s{}^*) \cdot \left(\frac{0.033}{s_{ax}{}^*}\right)^{0.9} \cdot \left(\frac{0.02}{g_a{}^*}\right)^{-0.75}$ Limit m_c* to a maximum of 3	
Definitions	$s_{ax}{}^* = \frac{s_{ax}}{r_2}; \; L_s{}^* = \frac{L_s}{r_2}; \; g_a{}^* = \frac{g_a}{r_2}$	

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NOTE. When using these correlations, it is advisable not to extrapolate beyond above limits but use the cut-off values instead as given by above maxima and minima.

NOTE. The correlations are also subject to large uncertainties. Only the correlation for the direct stiffness looks quite good (R^2 = 0.93). The correlation coefficient of c* was R^2 = 0.89 while all other coefficients correlated poorly (R^2 = 0.63 to 0.7).

The dimensioned coefficients are obtained as shown in Table 4.

Table 4. Hydraulic impeller interaction coefficients.

Direct stiffness	$k = X_F \cdot \omega^2 \cdot k^*$	$X_F = \pi \cdot \rho \cdot r_2^2 \cdot b_2$
Cross-coupled stiffness	$k_c = X_F \cdot \omega^2 \cdot {k_c}^*$	
Direct damping	$c = X_F \cdot \omega \cdot c^*$	
Croos-coupled damping	$c_c = X_F \cdot \omega \cdot {c_c}^*$	
Direct mass	$m = X_F \cdot m^*$	
Cross-coupled mass	$m_c = X_F \cdot {m_c}^*$	

Refer to Mod.494 – Hydraulic Impeller Interaction to define hydraulic impeller interaction coefficients for the different rotating speeds.

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5.4. Fluid added mass

The fluid surrounding the rotor adds inertia to the rotor in three ways:

- the fluid trapped in the impeller passages adds mass directly, and this can be calculated based on the volume in the impeller passages times the pumped fluid density.
- the fluid around the periphery of the impellers that is displaced by the vibrating motion of the impellers. This is discussed in the literature (Blevins 1984), which show how this part of the added mass is equal to the "swept volume" of the impellers and immersed shafting, times the density of the pumped liquid (Figure 10). Depending upon impeller type based on the pump specific speed, this second form of added mass can be a significant percentage of the impeller metal mass.
- the fluid in close clearances can be significant. for high frequency vibration (such as in rocket turbopumps) or for long L/D passages (like in a canned motor pump) because it must accelerate to get out of the way of the vibrating rotor. (It is not the case in OH or BB pumps).

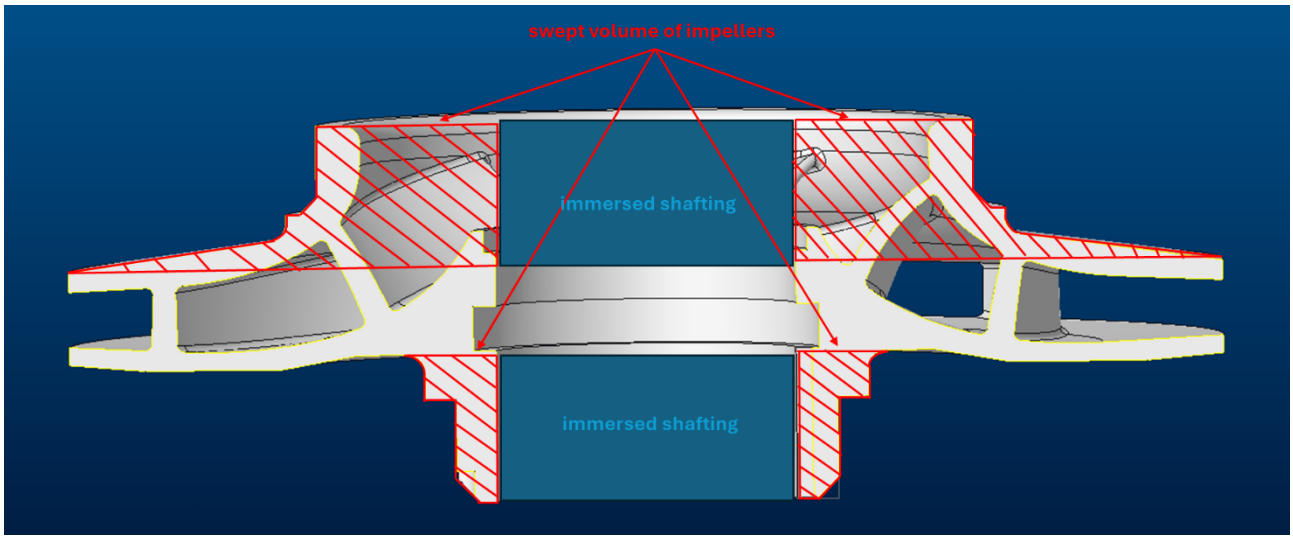

Figure 10. Fluid displaced by the motion of the impeller.

The fluid swept by immersed shafting can be estimated as:

- $m = \rho_f \cdot \frac{\pi}{4} \cdot (ODmb^2 - IDmb^2) \cdot L \quad [kg]$

The masses of trapped fluid and fluid swept by impellers can added to the impeller stations as concentrated mass.

NOTE. Do not include fluid contribution when performing dry analyses.

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6. Acceptance criteria.

As indicated in API 610 12th Edition Annex I,

- if the first bending natural frequency of “dry” analysis at separation margin is at least 20% above the highest potential excitation frequency (based on maximum continuous speed), no more analysis are required;
- otherwise, perform a damped natural frequency analysis as explained in "*5 Methodology, analysis and interpretation of results.”.* For both as-new and twice the as-new clearance, the damping factor versus serration margin between any bending natural frequency and the synchronous run line shall be within the acceptable region shown on Figure 1.

If the damping factor versus separation margin for a mode or modes is not acceptable:

- if possible, an iteration of the analysis should be performed such that changes can be made to correct the issue. They may involve a shaft diameter, impeller and/or coupling weight, bearing and/or wear ring geometry modification.
- the rotor’s damped response to unbalance shall be determined for the mode(s) in question (Level 3 Force Response Analysis).

For a robust design, it is suggested to work outside the non-acceptability region of API 610 Annex I because a response to unbalance does not take into account other possible failure due to not-synchronous excitation forces.

Figure 11 summarizes the lateral analysis procedure.

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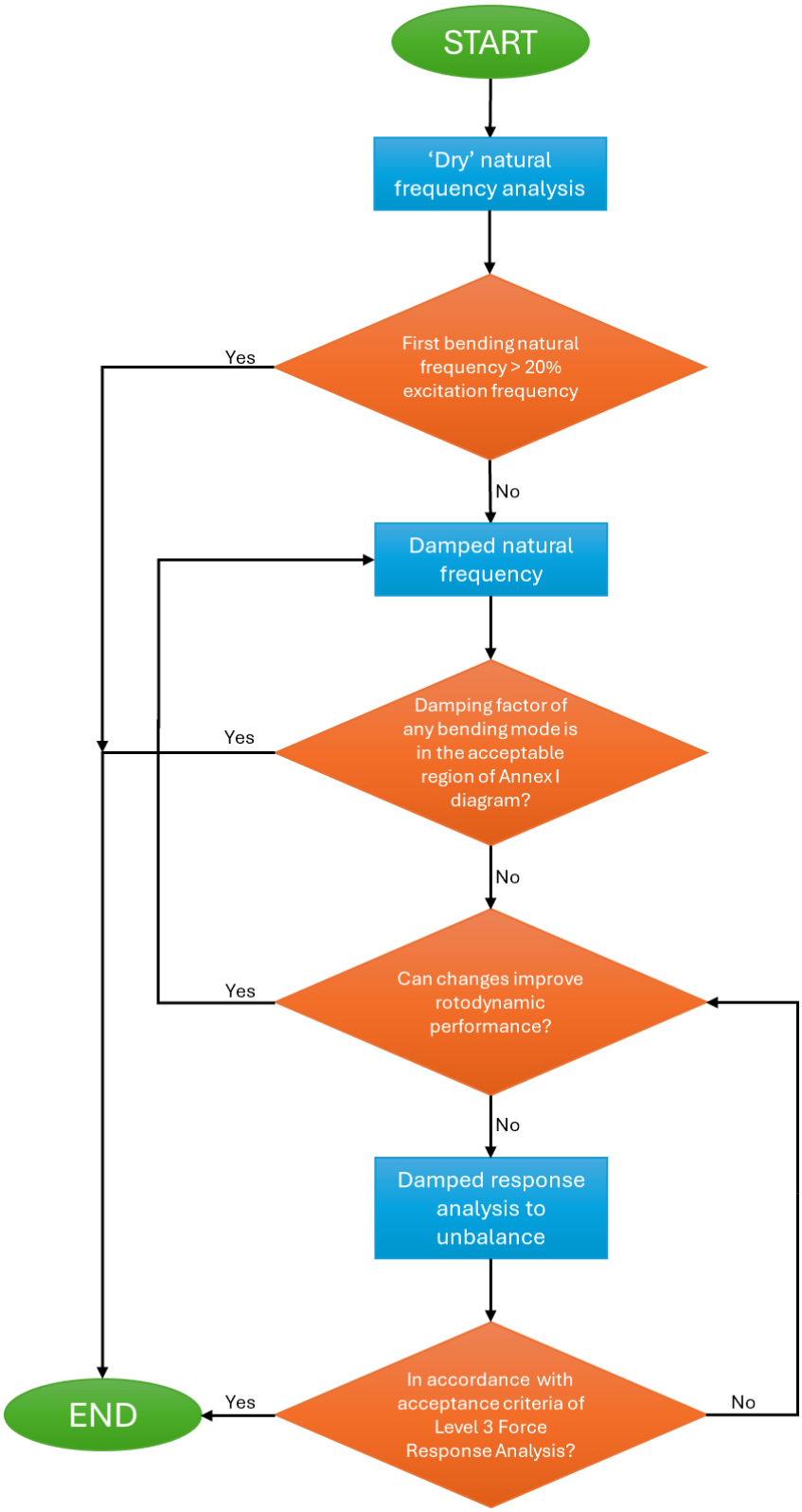

Figure 11. Lateral analysis flowchart.

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7. References

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G.10.00.09 - Guidelines for dynamic analysis complexity level

Mod.493 – Pressure drop across the seal

Mod.494 – Hydraulic impeller interaction coefficient

Mod.495 – Annex I Diagram

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Annex A. Input data requirements

The following information is typically required for a lateral rotordynamic analysis:

1. Pump shaft dimensions (drawing preferred)
2. Rotor material of construction
   - Shaft
   - Impellers
   - Sleeved
   - Coupling
3. Mass and inertia (polar and transverse) for all rotating components
   - Impellers
   - Sleeves
   - Mechanical seal rotating components

Note: mass and inertia can be estimated from component geometry if necessary.

4. Assembly information
   - Axial location of all rotating components on the shaft
   - Diametral fit of all rotating components on the shaft (loose or press, and if press, how much diametral interference)
5. Coupling information (drawing preferred, and usually available from vendor)
   - Manufacturer, type and size
   - Details of spacer (length, mass and inertia)
6. Rolling element bearing information
   - Manufacturer, type and size
   - Estimated radial loads on bearing
   - Bearing stiffness in both transverse direction (can be calculated by bearing manufacturer)
7. Fluid-film bearing information
   - Lubricant manufacturer, ISO viscosity grade, and temperature
   - Bearing drawing and shaft/sleeve drawing
     - Active close-clearance length
     - Radial clearance (new and worn)
   - Alternatively, bearing stiffness, and damping coefficients at operating conditions
8. Interstage seal/wearing ring/balance device information

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- Seal/wear ring/ balance device drawings
    - Active close clearance length
    - Grooving
    - Radial clearance
- Pumped fluid type and temperature (or fluid viscosity at operating temperature)
- Pressure developed per stage

9. Motor information (only needed if coupling is rigid)

- Rotor drawing
- Rotor mass and inertia (polar and transverse)
- Rotor core fit-up on shaft (loose or press, and if press, how much diametral interference)
- Motor fan mass and inertia (polar and transverse)
- Motor fan fit-up on shaft
- Motor bearing information (same information as above for pump bearings)

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Annex B. Lateral rotordynamic analysis with MADYN 2000.

1.1. Madyn 2000, System object

In MADYN 2000 the model, the loads and further analysis parameters as well as analysis results are stored in the system object. The model can consist of several objects such as shafts, couplings, bearings, etc...

A vibration analysis consists of the following steps:

- Modelling a system,
- defining loads on the system,
- running the analyses,
- plotting, listing and interpretation of results.

Each of these steps can be controlled from the system explorer.

The modelling process can be started from the “File” menu of the main MADYN 2000 window by the menu item “Create new System” (Figure 12).

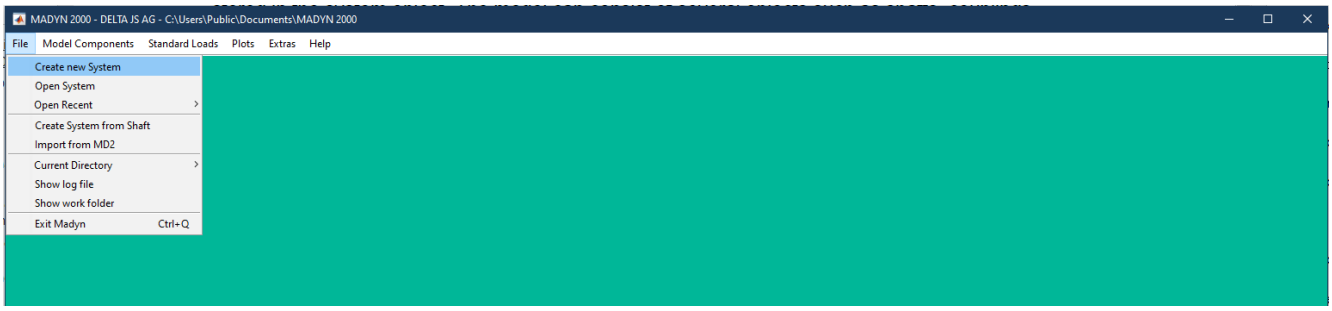

Figure 12. Creation of a new System.

The “Create new System” item leads to the opening of the *system explorer* and the *system GUI* (Figure 13). The *system GUI* has buttons to add shafts, gears and flexible couplings. The connection button opens a GUI to connect shafts and couplings. The pull-down menu “TLA case” allows to select the type of analysis (lateral, torsional, …).

The “add …” buttons in the *system GUI* open the “Cancel”/ “Create New”/ “File” window to either create a new object via its GUI or to select a file with the corresponding object (shaft, gear or flexible coupling) to be loaded.

Usually, shaft objects are uploaded from a file.txt following the procedure in *1.2 Madyn 2000, Shaft object*.

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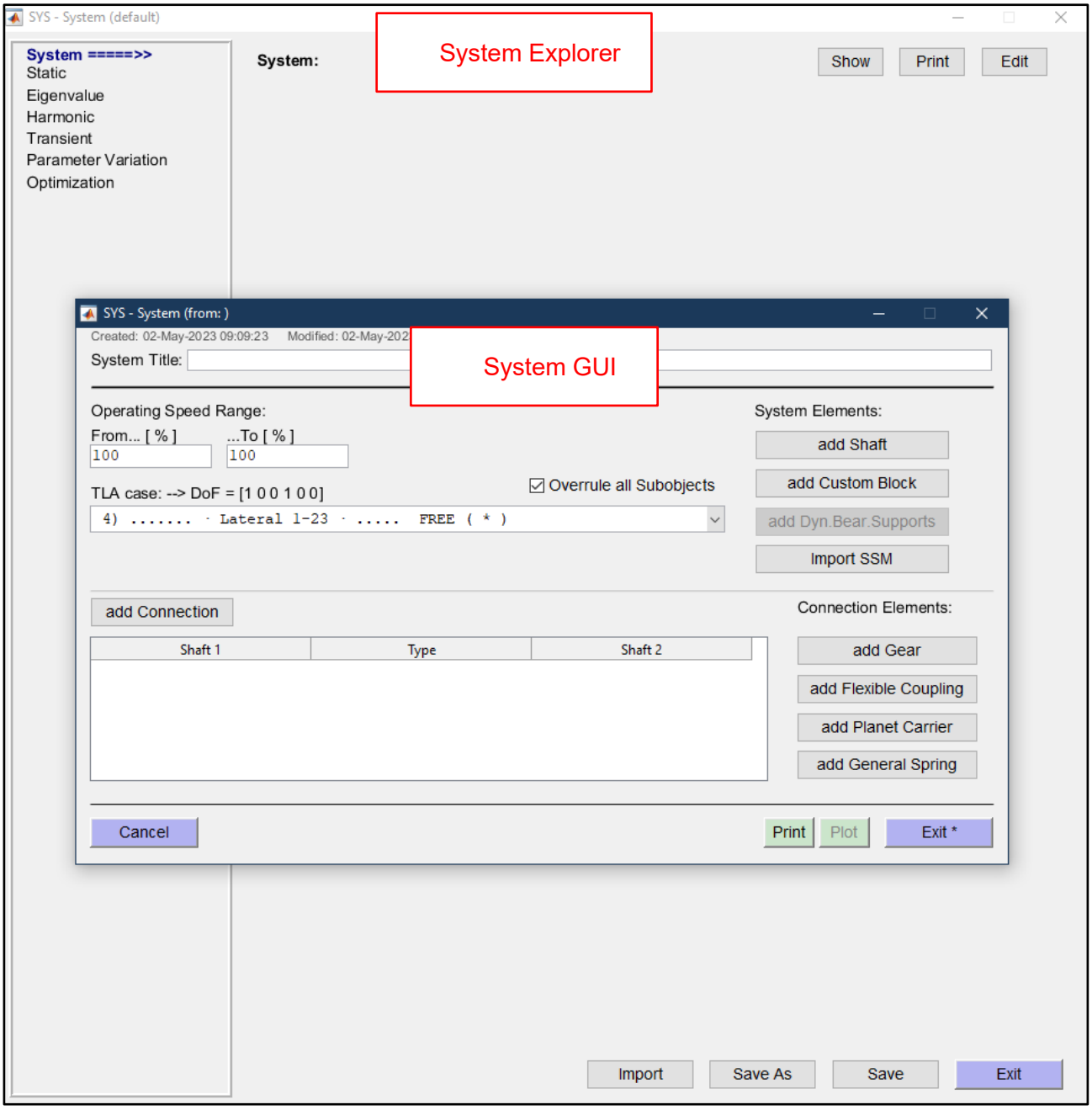

Figure 13. System explorer and system GUI.

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Figure 14 is an example of a *system GUI* for lateral analysis:

- The operating speed range goes form 25% to 125% as recommended in API 610 Annex I.1.2.
- TLA case is set on lateral analysis.
- Shafts constituting the train are uploaded.
- Connection elements are created (only if rigid couplings are present).
- Connections between shafts are defined (only if rigid couplings are present).

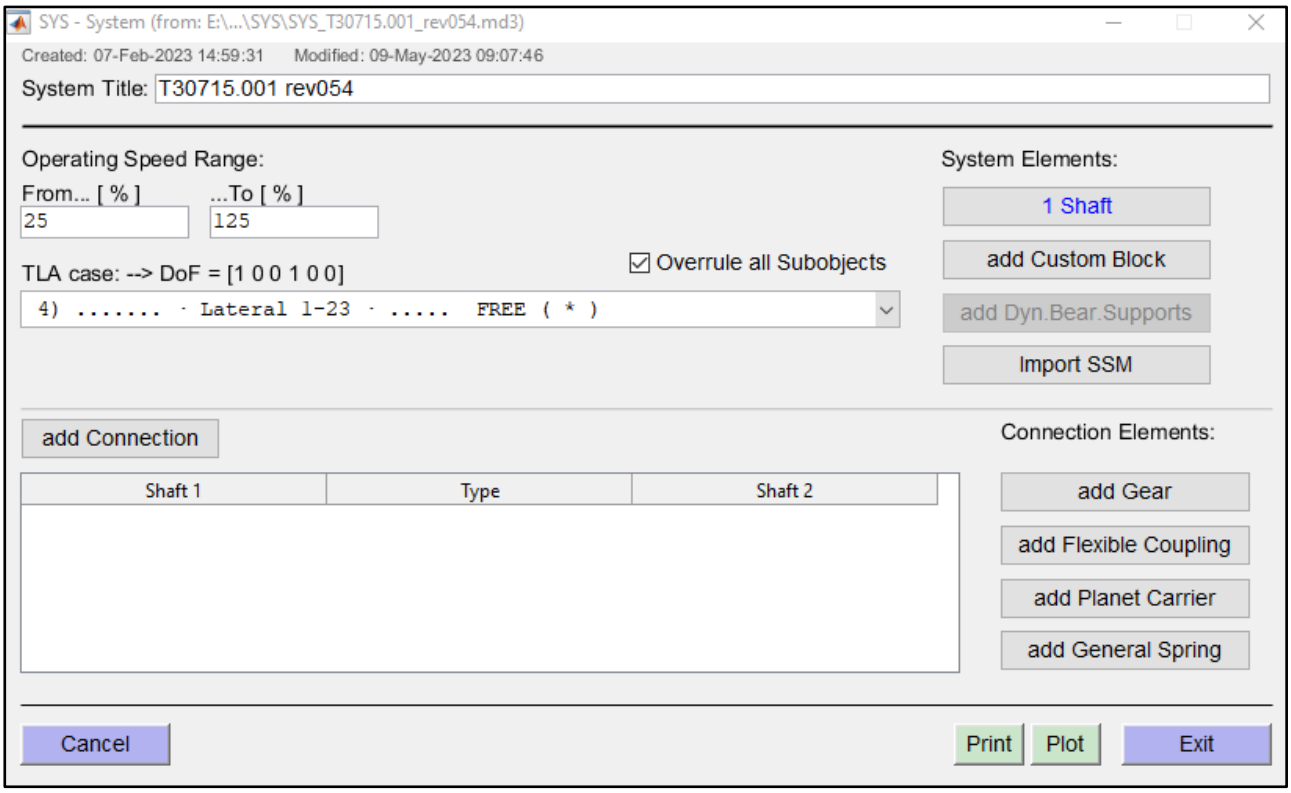

Figure 14. Example of System GUI.

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1.2. Madyn 2000, Shaft object

The creation of a rotor model starts from a spreadsheet that can be found in the directory *C:\Users\Public\Documents\MADYN 2000\Examples\Shaft_Input_w_EXCEL.*

The spreadsheet has different columns that need to be compiled.

- Columns “Length” and “ODsb”, that stands for outer bending stiffness diameter, are the mainly used. Each element should have a ratio L/D less than 1 with recommended ratio 0.5 (for example, length defined in Figure 15 is not correct).
- If there is a hollow section, the internal diameter should be defined in column “IDsb”, that stands for inner bending stiffness diameter.
- If there is a change in diameter in an element, the value in the “Conical” column needs to be changed to 1. 0 stands for a cylindrical element, while 1 for a conical one. To define a conical section, it is needed of two rows. In the first one the left-side diameter will be put, while in the other one the right-side diameter.
- If there are parts in the rotor that gives mass contribution but do not add stiffness (such as sleeve), the values in column “ODmb” must be changed to the outer diameter of this part. For default, these values are equal to “ODsb”. The same argument counts for the inner diameter: the values in column “IDmb” must be changed.
- Stations should be defined where there is the center of gravity of mounted parts on shaft, such as impellers, couplings bearing, wear rings, …
  Lumped masses and moment of inertia can be inserted through columns “M”, “Thetap” and “Thetae”. Column “Centre” defines the axial displacement of the lumped mass from the station assigned to the corresponding value.
- Columns “addmass”, “addinertiap” and “addinertiab” defines mass and moments of inertia per unit length. “addmass” can be used to define the fluid swept by the rotor during operation.
- Column “Denotation” is useful to highlight station of relevance. It will be helpful later in the model to locate these stations.

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Shaft Model Data for MADYN 2000 14/04/2023

Disclaimer:
No responsibility for the correctness of the calculations in this spreadsheet is taken by DELTA JS AG.

Shaft Summary:		
Density:	7'850	[kg/m3]
Length:	2'720.5	[mm]
Mass:	2'687.3	[kg]
Thetap:	201.28	[kgm2]

Auxiliary values needed for the calculation of total shaft mass and polar moment of inertia:

Shaft Title

* #	Conical	Length	IDsb	IDmb	IDst	ODsb	ODmb	ODst	M	Thetap	Thetae	Centre	addmass	addinertiap	addinertiab	Temperature	Denotation	Sec. M	SEC Thetap	Add M	Add Thetap
*	[-]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]	[kgm2]	[kgm2]	[mm]	[kg/m]	[kgm2/m]	[kgm2/m]	[°C]	[-]	[kg]	[kgm2]	[kg]	[kgm2]
1	0	166.7	0.0	0.0	0.0	130.0	130.0	130.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		17.37	0.037	0.00	0.00
2	0	23.3	0.0	0.0	0.0	130.0	130.0	130.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		2.43	0.005	0.00	0.00
3	0	60.0	0.0	0.0	0.0	130.0	130.0	130.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40	'Coupling connection'	6.25	0.013	0.00	0.00
4	0	200.5	0.0	0.0	0.0	140.0	140.0	140.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		24.23	0.059	0.00	0.00
5	0	15.0	0.0	0.0	0.0	156.0	156.0	156.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		2.25	0.007	0.00	0.00
6	0	45.0	0.0	0.0	0.0	172.5	172.5	172.5	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		8.26	0.031	0.00	0.00
7	0	44.5	0.0	0.0	0.0	172.5	172.5	172.5	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40	'DE bearing'	8.16	0.030	0.00	0.00
8	0	73.0	0.0	0.0	0.0	200.0	200.0	200.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		18.00	0.090	0.00	0.00
9	0	226.4	0.0	0.0	0.0	247.4	247.4	247.4	50.0	3.00	1.50	-100.00	1500.0	150.0	75.0	40	'End windings'	85.43	0.654	339.60	33.96
10	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		77.38	0.592	307.58	30.76
11	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		77.38	0.592	307.58	30.76
12	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40	'Centre of lamination'	77.38	0.592	307.58	30.76
13	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		77.38	0.592	307.58	30.76
14	0	226.4	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		85.43	0.654	339.60	33.96
15	0	147.0	30.0	30.0	30.0	200.0	200.0	200.0	50.0	3.00	1.50	100.00	0.0	0.0	0.0	40	'End windings'	35.44	0.181	0.00	0.00
16	0	50.0	30.0	30.0	30.0	172.5	172.5	172.5	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		8.90	0.034	0.00	0.00
17	0	82.0	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		10.77	0.031	0.00	0.00
18	0	136.5	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		17.93	0.052	0.00	0.00
19	0	60.0	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		7.88	0.023	0.00	0.00
20	0	32.0	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		4.20	0.012	0.00	0.00
21	0	44.0	30.0	30.0	30.0	131.0	131.0	131.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		4.41	0.010	0.00	0.00
22	0	44.0	30.0	30.0	30.0	131.0	131.0	131.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40	'NDE bearing'	4.41	0.010	0.00	0.00
23	1	140.0	30.0	30.0	30.0	131.0	131.0	131.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		12.82	0.027	0.00	0.00
23	1	0.0	30.0	30.0	30.0	120.0	120.0	120.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		0.00	0.000	0.00	0.00
24	0	84.0	30.0	30.0	30.0	90.0	90.0	90.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		3.73	0.004	0.00	0.00
25	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
26	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
27	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
28	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
29	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
30	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00

Figure 15. Example of spreadsheet for rotor input data.

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After compiling the spreadsheet, the part including rotor geometry should be copied and saved as a file.txt. It is important to use a proper name for the file because Madyn 2000 recognize the file as a shaft object only if the file name has a prefix SFT_ (i.e. SFT_example).

SFT_shaft.txt - Blocco note di Windows

File Modifica Formato Visualizza ?

```
Shaft Title
* #     Conical Length  IDsb   IDmb   IDst   ODsb   ODmb   ODst   M      Thetap  Thetae  Centre  addmass addinertiap     addinertiab     Temperature     Denotation
*       [-]     [mm]    [mm]   [mm]   [mm]   [mm]   [mm]   [mm]   [kg]   [kgm2]  [kgm2]  [mm]    [kg/m]  [kgm2/m]       [kgm2/m]        [°C]    [-]
1       0       166.7   0.0    0.0    0.0    130.0  130.0  130.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
2       0       23.3    0.0    0.0    0.0    130.0  130.0  130.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
3       0       60.0    0.0    0.0    0.0    130.0  130.0  130.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40      'Coupling connection'
4       0       200.5   0.0    0.0    0.0    140.0  140.0  140.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
5       0       15.0    0.0    0.0    0.0    156.0  156.0  156.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
6       0       45.0    0.0    0.0    0.0    172.5  172.5  172.5  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
7       0       44.5    0.0    0.0    0.0    172.5  172.5  172.5  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40      'DE bearing'
8       0       73.0    0.0    0.0    0.0    200.0  200.0  200.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
9       0       226.4   0.0    0.0    0.0    247.4  247.4  247.4  50.0   3.00    1.50    -100.00 1500.0  150.0   75.0    40      'End windings'
10      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0    40
11      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0    40
12      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0    40      'Centre of lamination'
13      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0    40
14      0       226.4   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0    40
15      0       147.0   30.0   30.0   30.0   200.0  200.0  200.0  50.0   3.00    1.50    100.00  0.0     0.0     0.0     40      'End windings'
16      0       50.0    30.0   30.0   30.0   172.5  172.5  172.5  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
17      0       82.0    30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
18      0       136.5   30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
19      0       60.0    30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
20      0       32.0    30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
21      0       44.0    30.0   30.0   30.0   131.0  131.0  131.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
22      0       44.0    30.0   30.0   30.0   131.0  131.0  131.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40      'NDE bearing'
23      1       140.0   30.0   30.0   30.0   131.0  131.0  131.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
23      1       0.0     30.0   30.0   30.0   120.0  120.0  120.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
24      0       84.0    30.0   30.0   30.0   90.0   90.0   90.0   0.0    0.00    0.00    0.00    0.0     0.0     0.0     40
```

Figure 16. Example of SFT_file.txt

In order to create the shaft object from the SFT file, it is necessary to follow the path in Figure 17.

If it is the first time that an SFT file will be imported, before importing the file, it is necessary to upload in Madyn the SIF_Shaft.txt file from the command “SFT – Ask for SIF and Load”. The folder containing this file is the same of the excel file:

- *C:\Users\Public\Documents\MADYN 2000\Examples\Shaft_Input_w_EXCEL.*

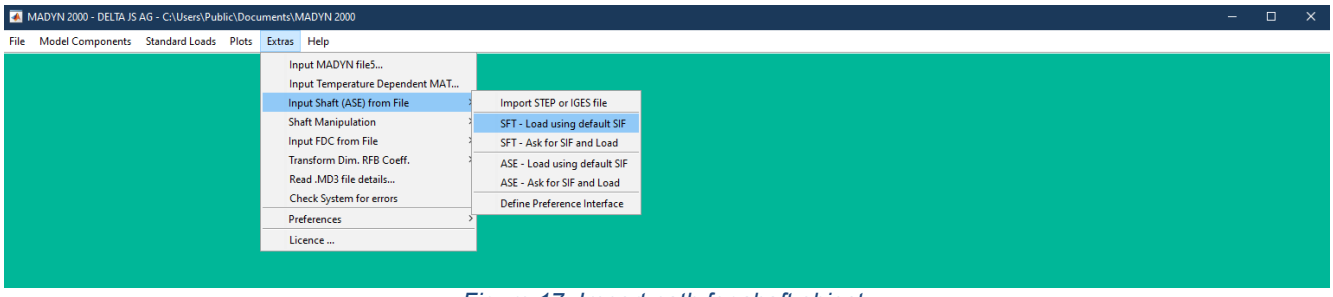

Figure 17. Import path for shaft object.

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Selecting the correct SFT file to open, Madyn 2000 generates the shaft object that can be modified through the interface in Figure 18.

The *shaft GUI* contains default values for the material and TLA case. Assign the appropriate material properties taking into account the influence of temperature on mechanical properties as indicated in “5.1 5.1Modelling shaft”. It is important to define the correct rotating speed of the shaft.

The shaft object can be saved through right click or with the Exit command.

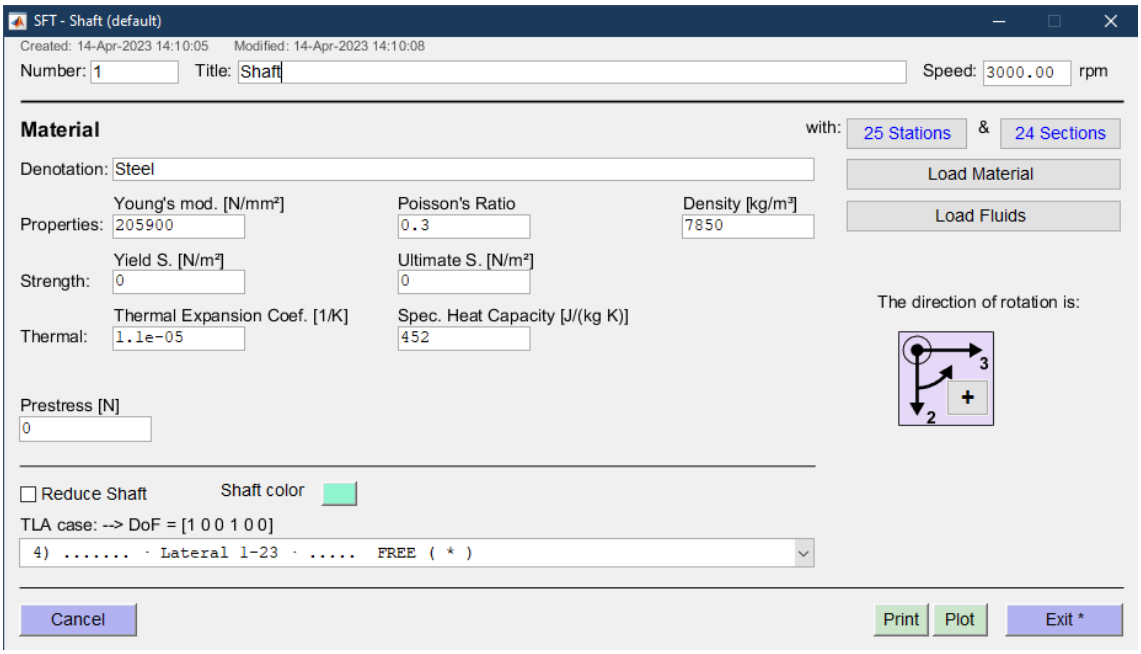

Figure 18. Shaft GUI.

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1.3. Madyn 2000, FDC object

FDC objects are characterised by the 6 following coefficients, which can be speed dependent:

- k, direct stiffness coefficient
- k_q, cross coupling stiffness coefficient
- d, direct damping coefficient
- d_q, cross coupling stiffness coefficient
- m, direct mass coefficient
- m_q, cross coupling mass coefficient

FDC forces are defined as follows:

$$-\begin{bmatrix} F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} k & k_q \\ -k_q & k \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} d & d_q \\ -d_q & d \end{bmatrix} \begin{bmatrix} \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} + \begin{bmatrix} m & m_q \\ -m_q & m \end{bmatrix} \begin{bmatrix} \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix}$$

There are two option to receive FDC coefficients in Madyn 2000:

- Import coefficients
- Define the geometry, fluid and operating conditions and the calculate the coefficients

These options are available in the pull down menu “Origin” of the FDC Gui under “Imported” and “User Defined”.

Imported coefficients (Figure 19). The coefficients can be defined in edit fields. For the definition of speed dependent fluid the check box “Speed dependent fluid interpolation” must be active. Speeds can then be defined in an edit field and for each speed coefficients can be input in the respective edit field. The number of coefficients must correspond to the number of speeds. Speeds are interpolated, but not extrapolated.

User defined coefficients. A GUI defines the geometry, the fluid and operational parameters such as pressures, inlet temperature, speeds and inlet swirl. The seal analysis can be carried out for compressible fluid (gases) and incompressible fluids (liquids, Figure 20).

Since it requires the absolute pressures at inlet and outlet, the inlet pressure can be defined as the pressure drop estimated plus the outlet pressure assigned.

There are different seal types available as seal geometry. The more used are:

- Plain annular seals;
- See through labyrinth with rotor or stator strips.

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For each of the seals specific geometry parameters can be defined, apart from the general parameters which are the same for all seal types.

It is always recommended to use coefficients including mass terms.

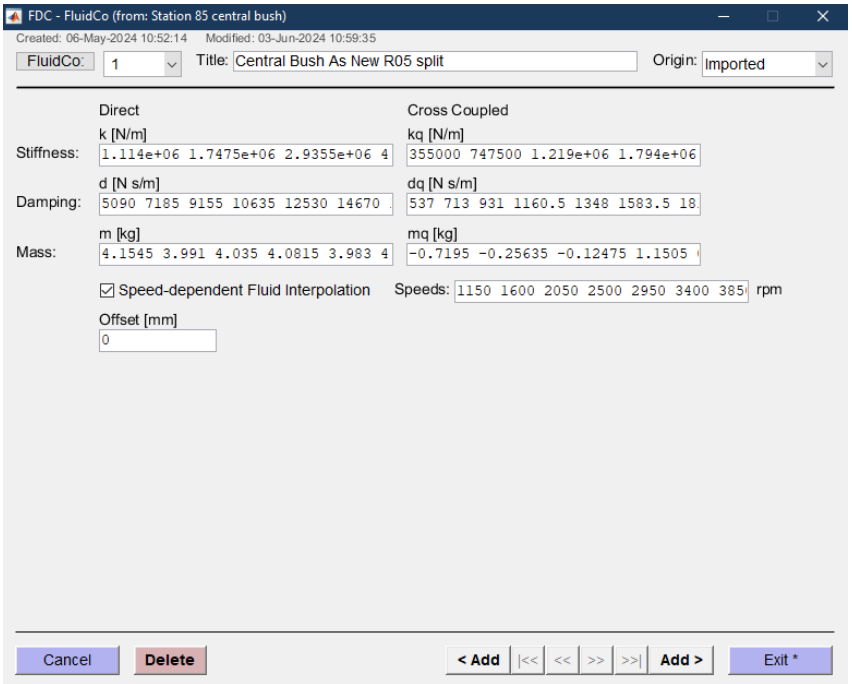

Figure 19. Imported FDC coefficients.

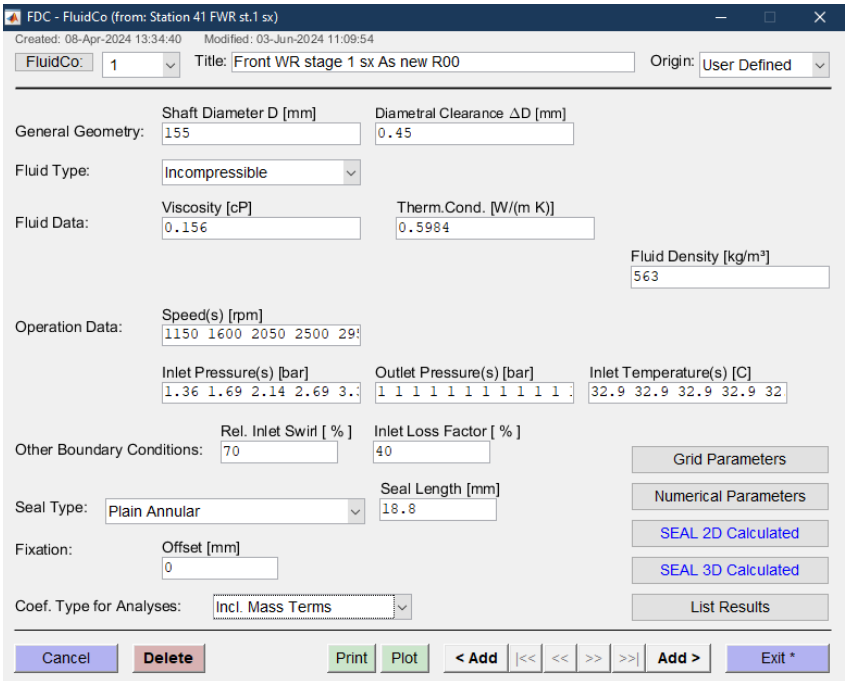

Figure 20. User defined incompressible FDC.

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1.3.1. Madyn 2000, Long seals

In Madyn 2000, FDC do not consider rotating degree of freedom. To include the tilting effect of the rotor, follow the procedure explained in “5.2.3 Long seals”.

To halve dynamic coefficients, refer to Mod.493 – Pressure Drop across the Seal. It requires as input the file.txt obtained from the full length FDC object as show in Figure 21.

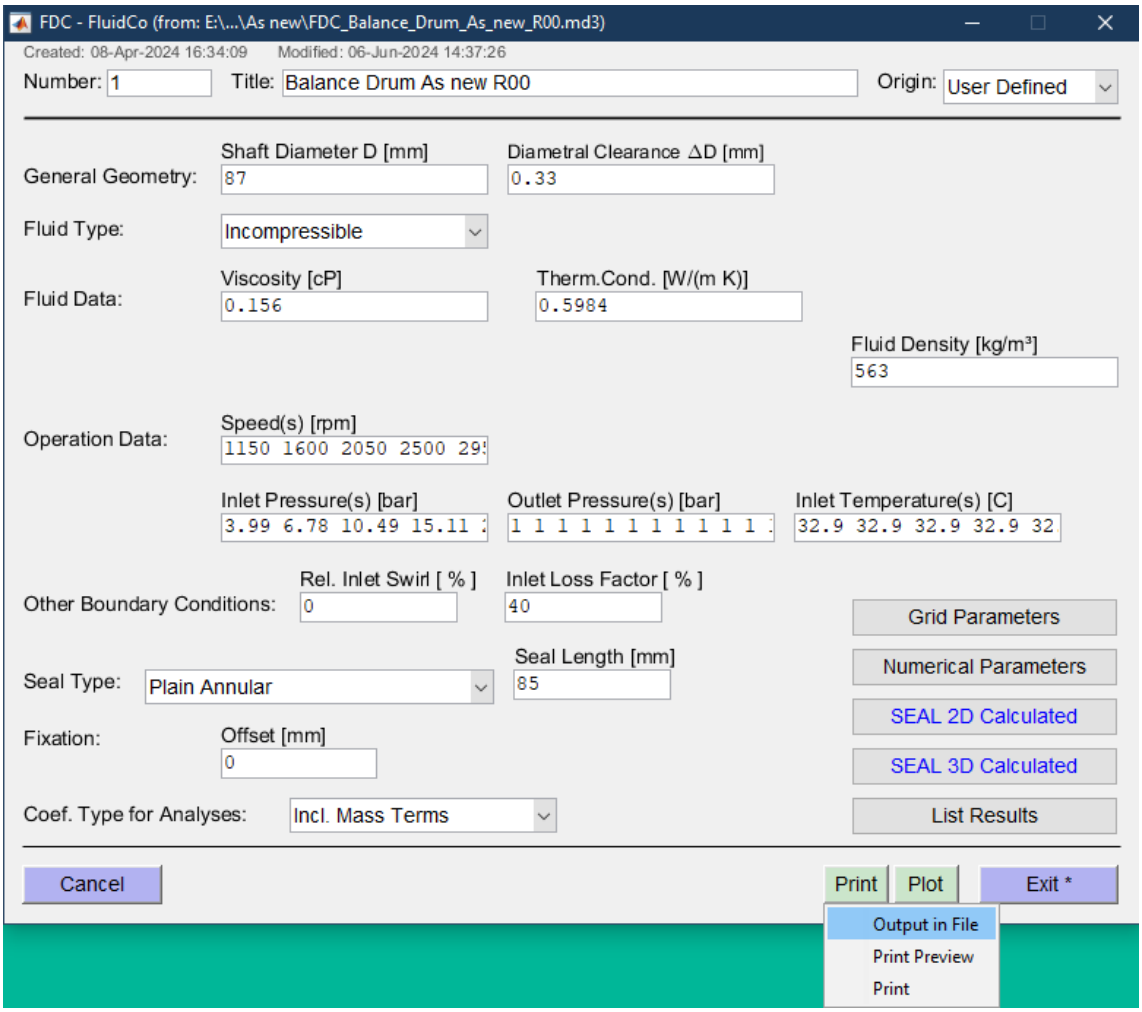

Figure 21. Export FDC object.

Mod.493 – Pressure Drop across the Seal generates as output a file.txt with the dynamic coefficients halved. This file should be used to create the new FDC object as shown in Figure 22.

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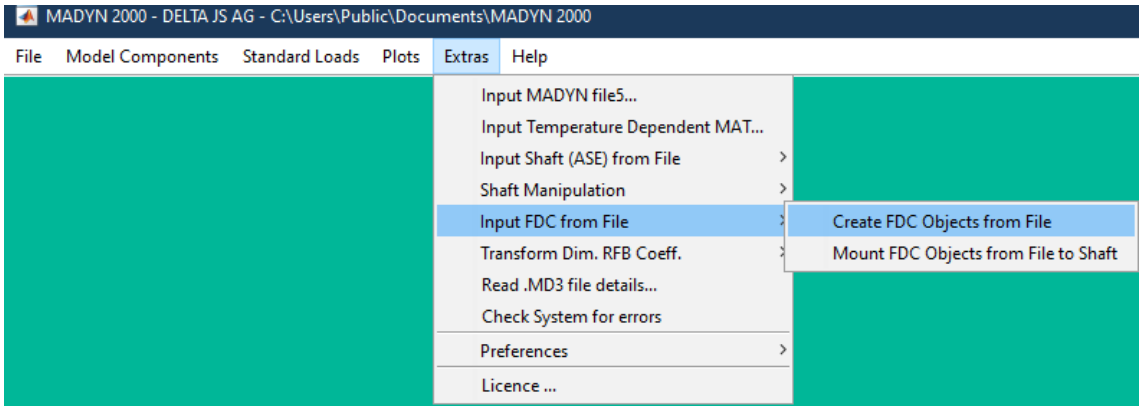

Figure 22. FDC Objects from File.

1.3.2. Madyn 2000, Hydraulic impeller interaction

The hydraulic impeller interaction coefficients, defined in “5.3 Hydraulic Impeller Interaction”, can be defined as Imported coefficient FDC object as described in 1.3 Madyn 2000, FDC object.

1.3.3. Product Lubricated bearing as FDC object

Product Lubricated bearing can be approximated as FDC objects when pressure drop and axial speeds across the seal are high. In case axial speed and pressure drop are low it should be better to model PLB as RFB objects (see 1.5 Madyn 2000, Fluid Film Bearings. RFB object).

When simulating PLB as FDC object, it is not possible to include axial grooves in the model. Only plain bearing can be studied.

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1.4. Madyn 2000, Rolling Element Bearings. REB object

With rolling element bearings, it is recommended to use axial lateral coupled system (Figure 23). The lateral behavior may be correct for the pure lateral system mostly in cases with 1 fixed and 1 loose bearing), however the axial vibrations, which can be excited by lateral forces, cannot be assessed.

Axial loads at the correct locations in the pump should be applied. The axially fixed bearings will take the loads and their properties will depend also on the axial load.

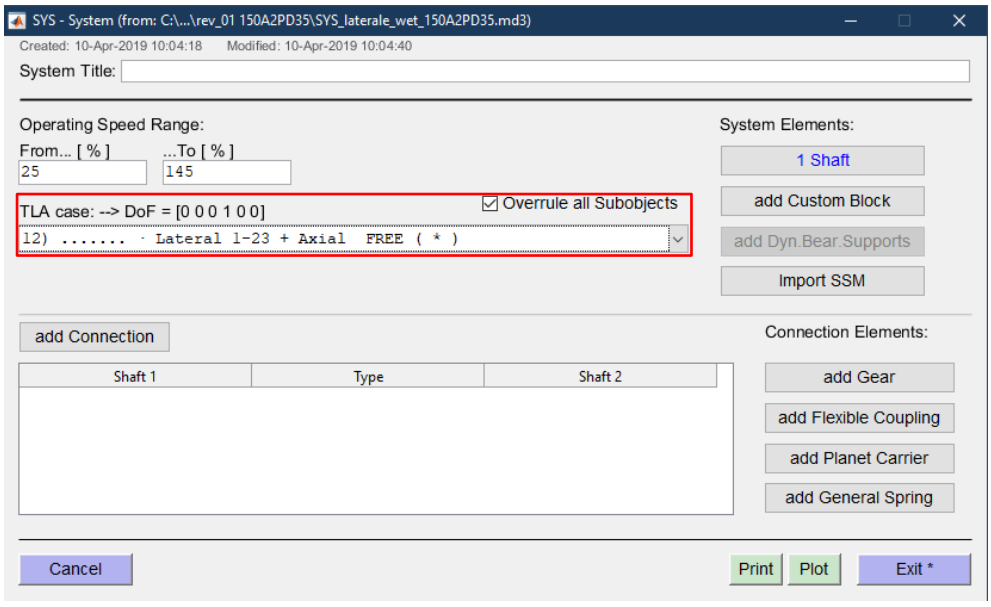

Figure 23. Axial and lateral coupled system.

For radial bearings the inner ring is connected. In case the outer ring rotates, a shaft in shaft connection must be used. The outer ring then is used as connector to the outer shaft.

In general rolling element bearings (REB) are speed dependent and the force displacement relation is nonlinear. The affected coordinates are the displacements in axial and radial direction x1, x2, x3 and rotations about the radial axes Φ2, Φ3. To calculate linear vibrations the force displacement relationship is linearized around the static equilibrium.

The GUI to define rolling element bearings is shown in Figure 24. In this GUI:

- a check box can be activated to consider the centrifugal force. Centrifugal forces will increase the pressure at the outer race but decrease the pressure at the inner race. It will lead to different contact angles on inner and outer race and an increased spin to roll ratio. When defining the REB, activate this option.

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- For bearing, which can take only radial loads select the boundary condition “Axially loose” while for those, which can take radial and axial loads select “Axially fixed”.
- A damping ratio D for harmonic response analysis can be defined. For this type of analysis, it is not required to define a value of damping ratio.

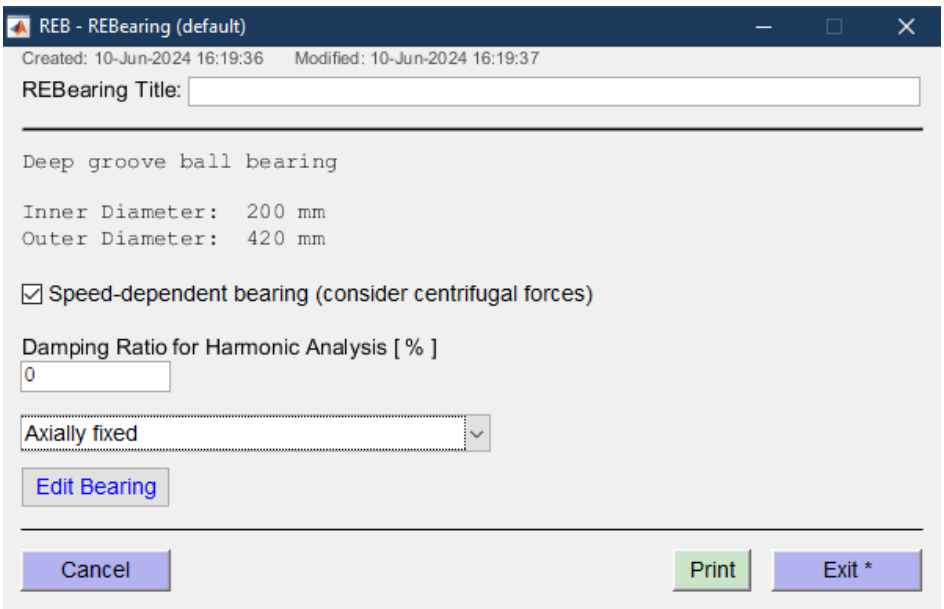

Figure 24. GUI Rolling Element Bearing.

The “Edit Bearing” button opens the GUI of MESYS (Figure 25) to define the bearing geometry, lubrication and material properties.

The MESYS software has a database to select among different sizes and types of ball and roller bearings.

In the “bearing geometry” window:

- Select the correct type of bearing (deep groove ball bearing, angular contact ball bearing, …);
- From the list chose the correct bearing.

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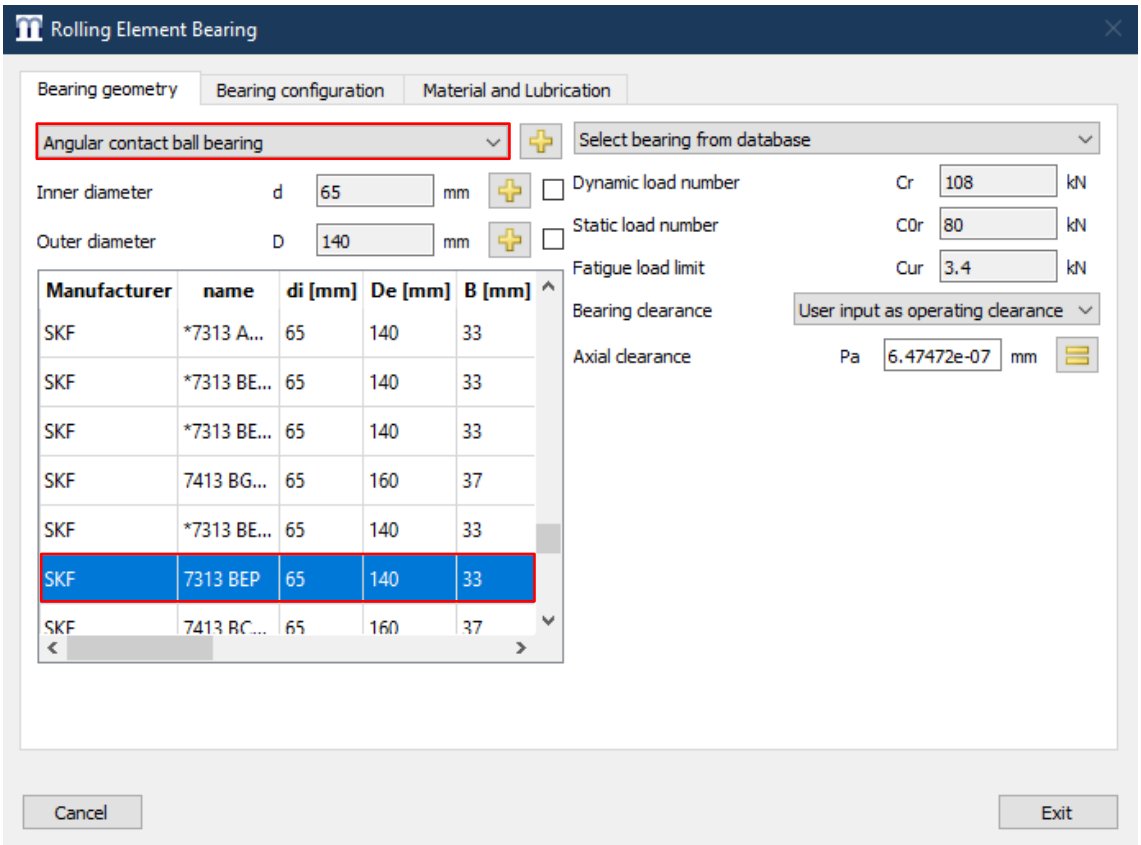

Figure 25. GUI Rolling Element Bearing.

1.4.1. Angular contact bearing: back-to-back configuration.

A group of bearings can be considered in the calculation. The bearings are assumed to be connected to a rigid shaft and housing. The loads or displacements are given for the origin of the bearing group instead of each single bearing.

A relative position can be entered for each bearing. An axial offset allows the definition of pretension in the system. The offset is an additional movement of the outer ring of a bearing. The displacement, forces and moments under loading are applied at the reference position 0. So, in most cases the position should be chose symmetrically around zero.

For bearings with a contact angle, the position of the cone center can be specified. In Figure 26, an O-configuration (or back-to-back configuration) is defined.

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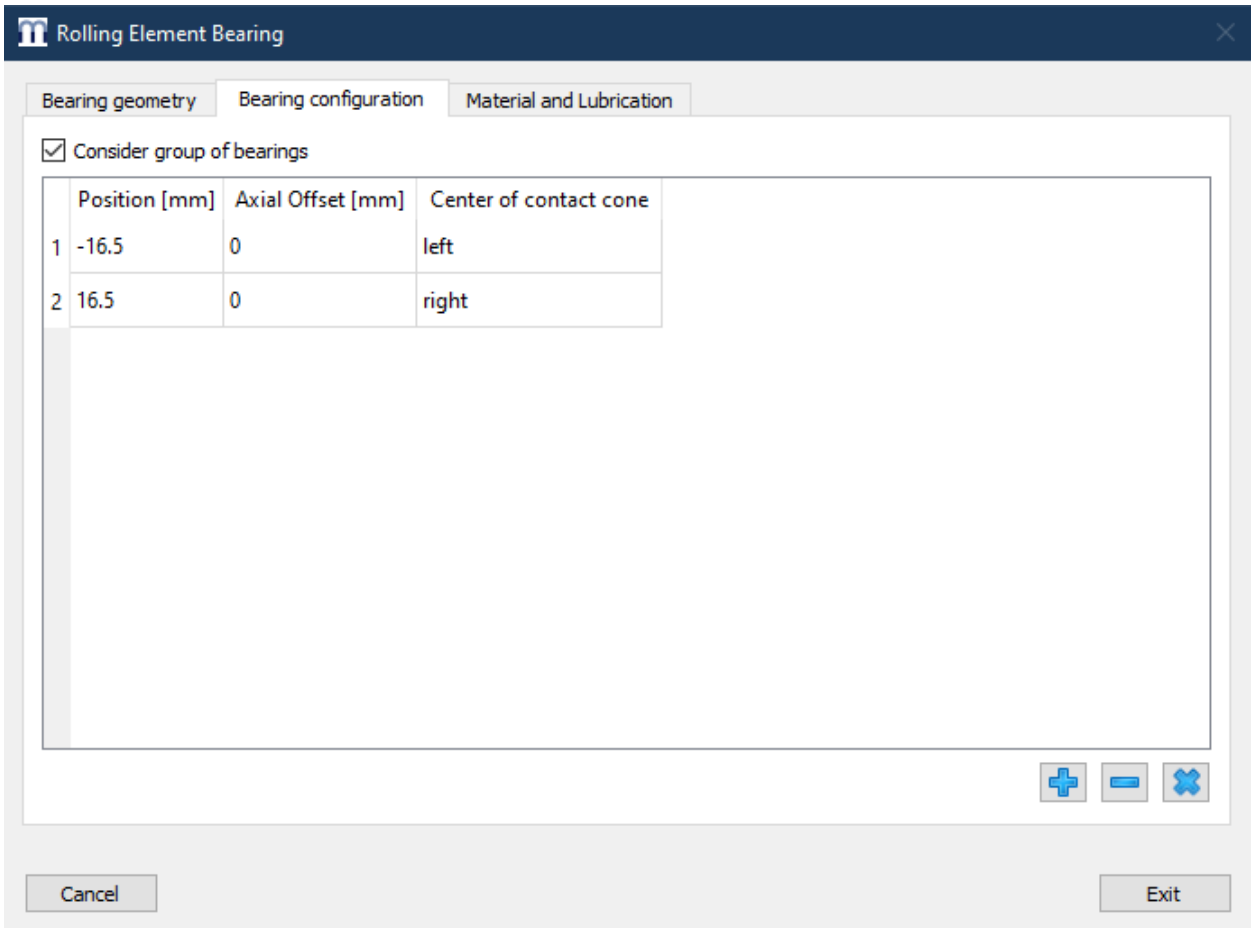

Figure 26. Back-to-back configuration.

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1.5. Madyn 2000, Fluid Film Bearings. RFB object

Radial journal bearings, as shown in Figure 27, can be modelled as RFB objects.

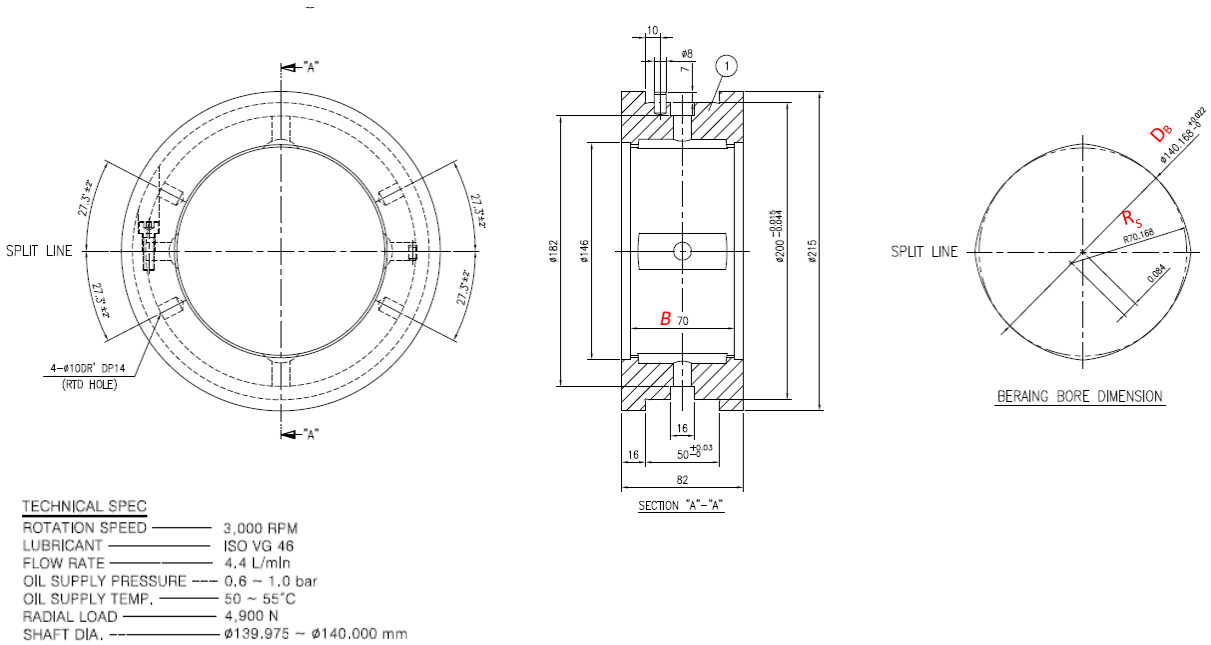

Figure 27. 4-lobe journals bearing.

- MADYN 2000 provides different ways to define a bearing. In this case, set "Origin" as User defined. The bearing geometry and the oil properties are completely defined in dimensioned form.
- In "Geometry" D is the shaft diameter. For a shaft diameter 140 mm h6, use the mean tolerated diameter (139.988 mm).

  B is the bearing width. In Figure 27 it is equal to 70 mm.

  Pad Type is Fixed.
- In "Clearance" ψ is the relative minimum clearance defined as defined as the ratio between the bearing radial clearance (dR) and shaft radius (D/2). The bearing radial clearance) is calculated as the difference between bearing radius ($D_B/2$) and shaft radius.

$$\psi = \frac{dR}{(D/2)}$$

$$dR = (D_B/2) - (D/2)$$

Preload m and pad clearance ratio ψ_v are correlated. It is sufficient to defined one of both. Pad clearance ratio ψ_v is equal to the ratio of the pad clearance (dS) and bearing radial clearance.

$$\psi_v = \frac{dS}{dR}$$

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The pad clearance dS is the difference between the pad radius (R_S) and the shaft radius.

$$dS = R_S - (D/2)$$

Preload m is equal to the ratio of the difference between pad clearance and bearing radial clearance and pad clearance.

$$m = \frac{(dS - dR)}{dS}$$

Use the mean values to estimate these parameters.

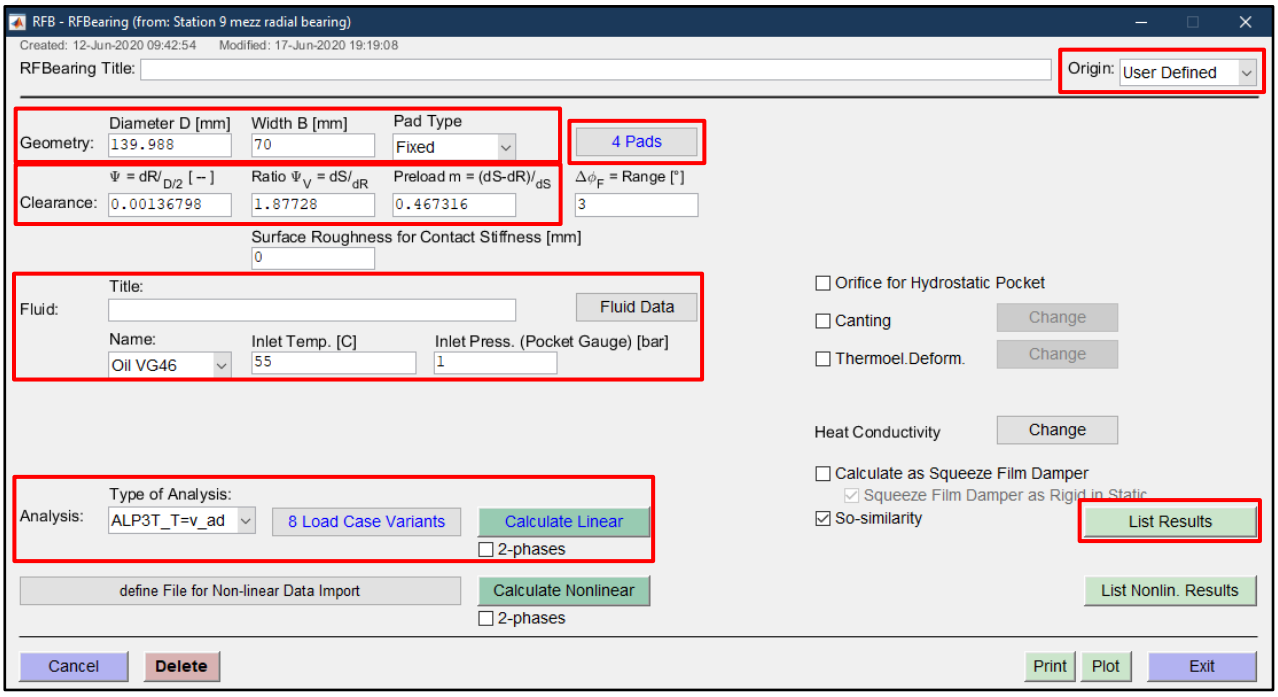

Figure 28. RFB object GUI: fixed lobe journal bearing.

- Click on the button “define Pads” to define the geometry of pads. Insert the number of pads. They will be 4 for the bearing in Figure 27.

For each pad you can redefine preload m and pad clearance ratio ψ_v by unchecking the option <u>common clearance</u>.

Angle sector, angle pad and curvature center define the circumferential position of each pad. It is suggested to include in each sector two half of each pocket bearing: one at the beginning between start angle sector and start angle pad and the other one between end angle pad and end angle sector. The curvature center should be middle sector angle plus 180° to have the minimum bearing radial clearance in the center of the sector. Command “Help*” will show an example how to define these angles.

Hydrostatic pockets are the area between pads where the lubricant is supplied. Pocket width will be defined relative to the bearing width. Pocket position 50% means, that it is

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axially centred. In case the width or the angle in circumferential direction are zero, the hydrostatic pocket is assumed as closed.

Define these parameters for each pad. To move between a pad and another use the arrows at the bottom.

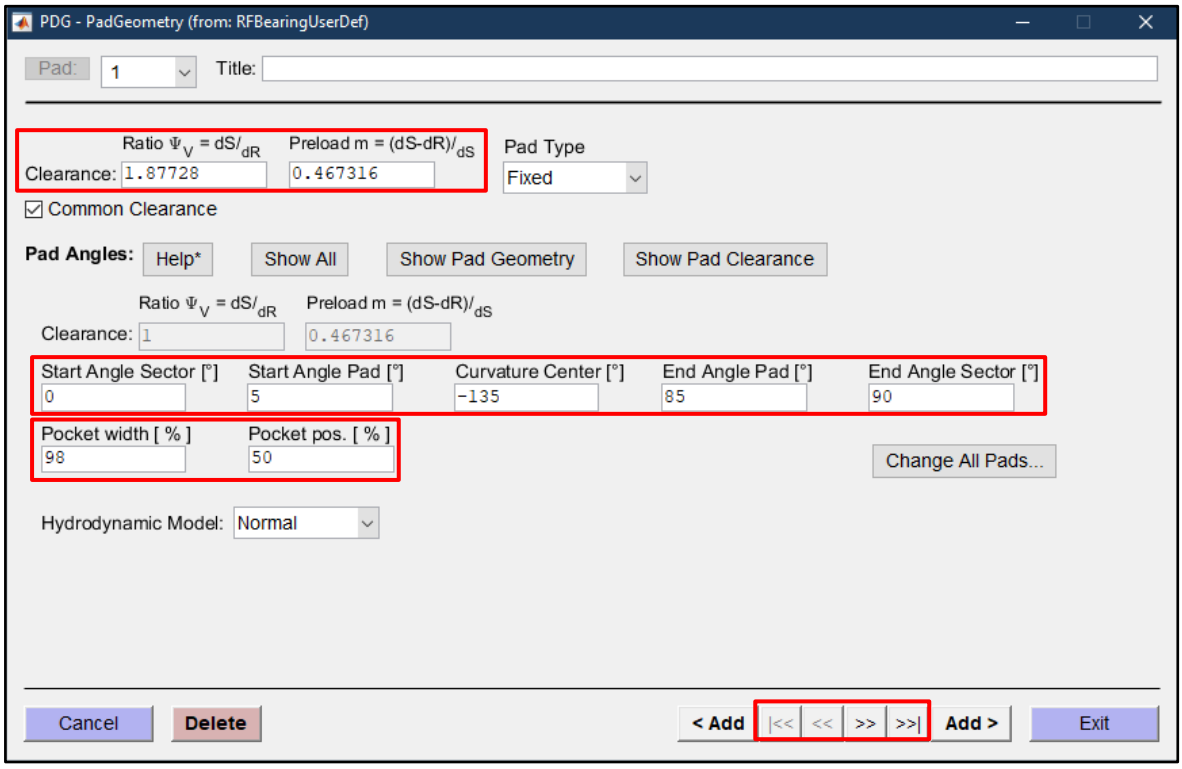

Figure 29. Pad Angles GUI: fixed lobe journal bearing.

- After defining pad geometry, RFB object requires oil properties (Figure 28). The fluid properties as a function of temperature are defined in tables for standard VG oils, water and Shell oil (VG32 and VG45). The temperature dependence of the viscosity plays a big role, especially for oil.

  In “Fluid” the type of oil, oil supply temperature and oil supply pressure (relative to atmospheric pressure) should be provided by the bearing supplier (Figure 27).
- In “Analysis” select as types of analysis: ALP3T_T=v_ad (variable adiabatic). The fluid film temperature is 3-dimensionally calculated with the assumption of an adiabatic bearing, i.e. the power loss is only dissipated by the oil. This analysis also takes into account turbulence. The Reynolds and Prandtl numbers are also dependencies apart from the Sommerfeld number for this analysis type. 2-phase flow for the cavitation zones can be considered.

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The analysis is carried out with ALP3T. In the analysis the journal position is varied until the pressure distribution yields a resulting force corresponding to the bearing load. The stiffness and damping coefficients are calculated from a perturbation of this equilibrium position.

In linear rotordynamic analyses coefficients can either be pre-calculated or automatically calculated when required in actual rotordynamic analyses such as harmonic response or Campbell diagrams. For the pre-calculation bearing loads and speeds must be defined in the bearing GUI (button “add Load Case Variant”). For the automatically calculated characteristics (suggested case) a load variant with adequate loads and speeds is created from the loads defined in the actual rotordynamic analyses. These loads are typically taken from a static analysis result.

- The results can be listed by the button “List Results”.

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Tilting pad journal bearing (Figure 30) can also be modelled using RFB objects.

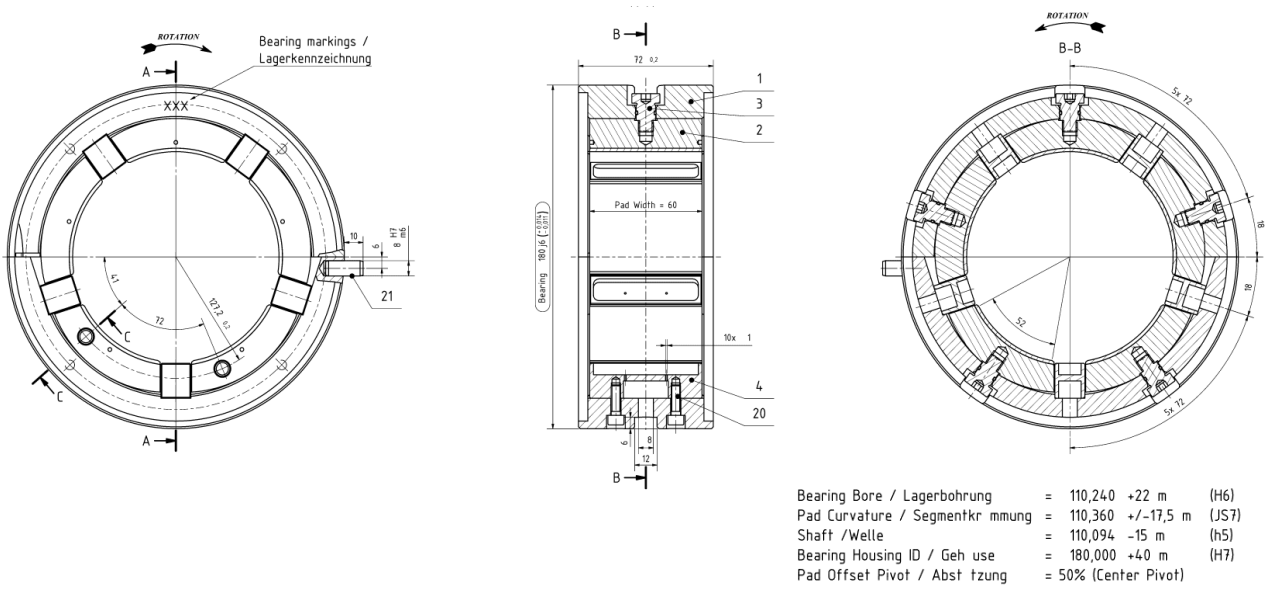

Figure 30.Titling pad journal bearing.

Settings are similar to the fixed journal bearing (Figure 31 and Figure 32). The main differences are:

- In “Geometry” set Pad Type as Tilting.
- In “define Pads” when defining the geometry of pads, instead of curvature center, set the Support Angle of the pad.
- In “Analysis” still select as <u>types of analysis</u>: ALP3T_T=v_ad (variable adiabatic) and activate the 2-phase flow option. When checking this option, two new options will appear in the “Fluid” section. The default settings go well.
- Set the “Oil Supply Condition” as Direct Lubrication. A Nozzle Area per Pad can be defined in the Pad Angle GUI. The inlet pressure then refers to the pressure before the nozzle/orifice, whereas otherwise it is the pressure in the hydrostatic pocket.
- Activate the Thermoelastic Deformation.

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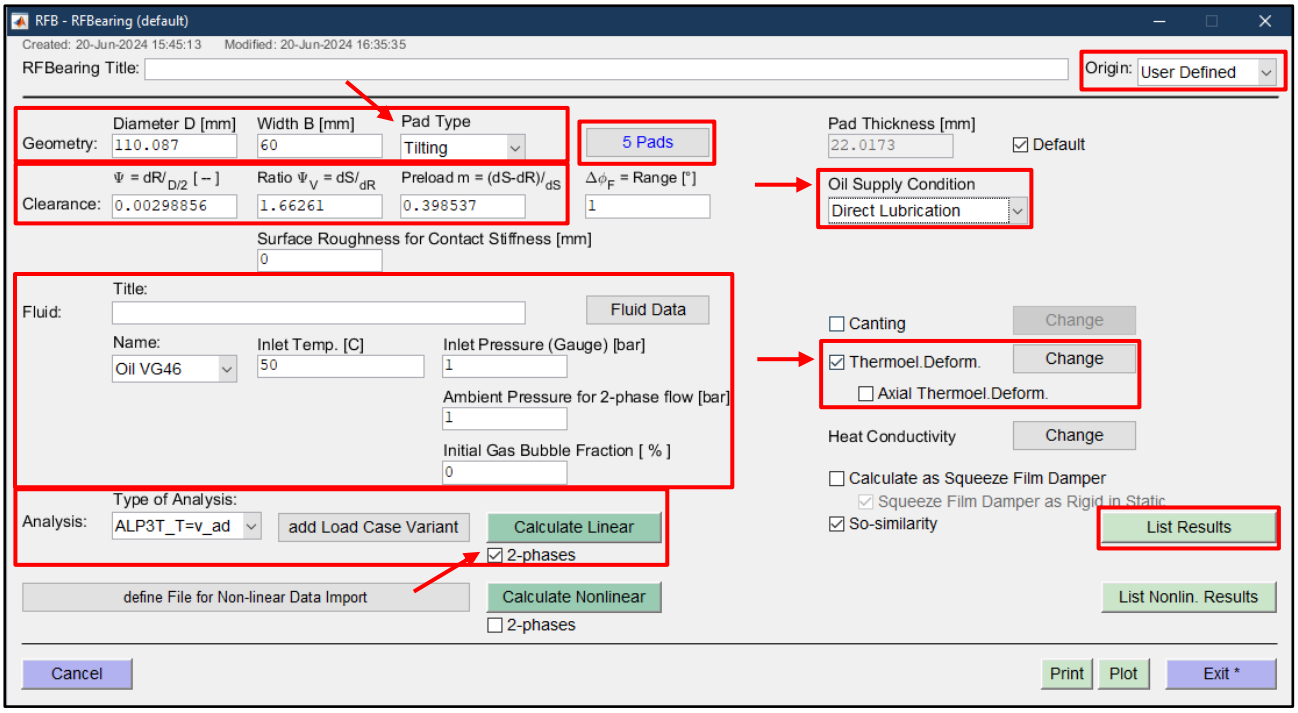

Figure 31. RFB Object GUI: Tilting pad journal bearing.

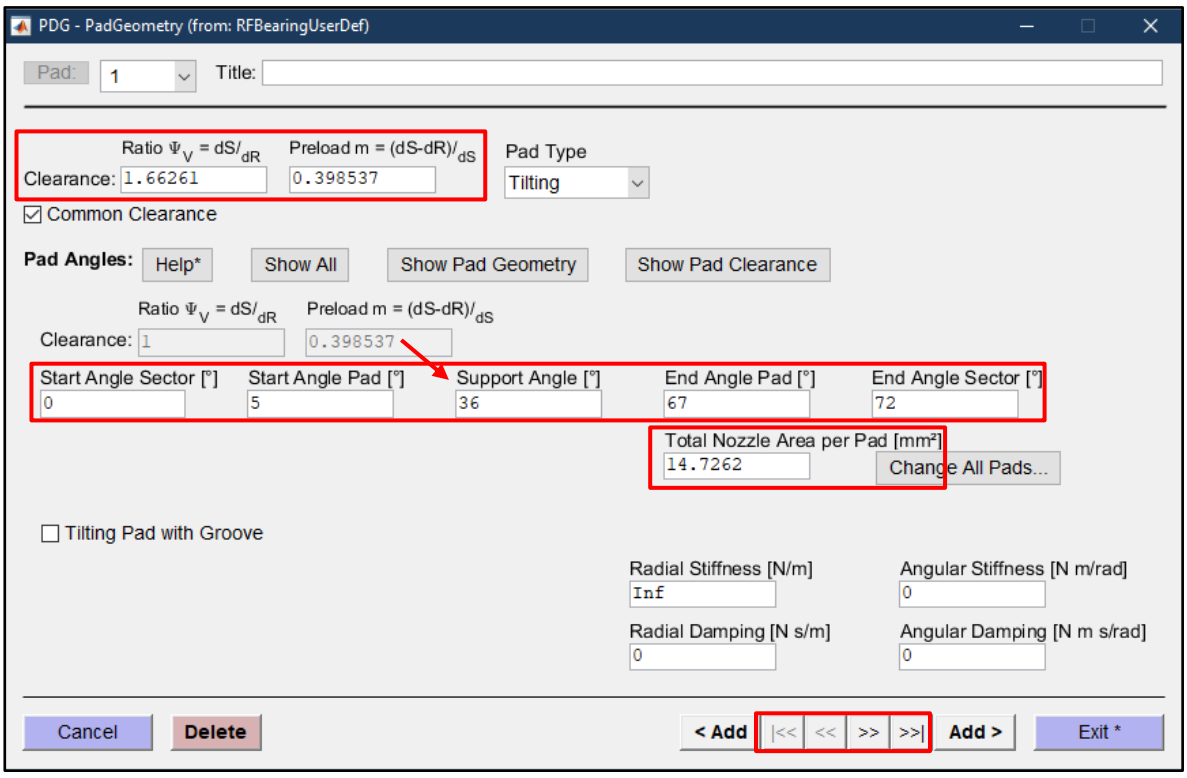

Figure 32. Pad Angles GUI: Tilting pad journal bearing.

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Product Lubricated bearing (PLB) can be approximated as RFB objects if the flow can be considered laminar (low pressure drop and low axial speed). In case axial speed and pressure drop are high it should be better to model PLB as FDC objects (see 1.3.2 Madyn 2000, Hydraulic impeller interaction

The hydraulic impeller interaction coefficients, defined in “5.3 Hydraulic Impeller Interaction”, can be defined as Imported coefficient FDC object as described in 1.3 Madyn 2000, FDC object.

Product Lubricated bearing as FDC object).

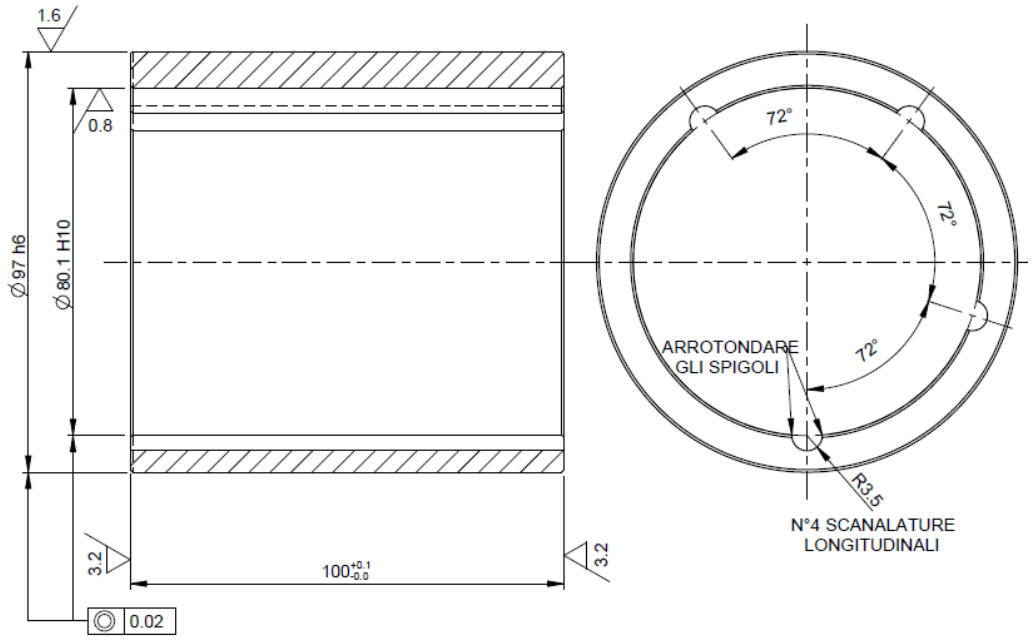

Figure 33. Product lubricated bearing.

- In “Geometry” define D (shaft diameter), B (bearing width) and Pad Type as Fixed.
- In “Clearance” set the relative minimum clearance ψ.

$$\psi = \frac{dR}{(D/2)}$$

$$dR = (D_B/2) - (D/2)$$

Preload m and pad clearance ratio ψ_v will be applied for each pad.

- Click on the button “define Pads” to define the geometry of pads. Insert the number of pads. They will be 8 for the bearing in Figure 33, 4 for the pads and 4 for the axial grooves.

Uncheck the option common clearance to set preload and pad clearance singularly. Apply a preload of 0 and a pad clearance ratio of 1 to the pad regions since they are generally

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plain. In these regions the curvature center is not important, use the middle sector angle (Figure 35).

In the axial groove regions (Figure 36), a pad clearance ratio should be defined as follow:

$$\psi_v = \frac{(s_{ax} + dR)/2}{dR}$$

Where s_{ax} is the equivalent depth of the groove. Calculate a depth that gives the same area of the groove defined. In Figure 33 the groove area can be approximated as:

$$S = (\pi \cdot R^2)/2 = (\pi \cdot 3.5^2)/2 = 19.25\ mm$$

Approximating the equivalent groove as rectangular with base 2R, the height will be equal to s_{ax} as follow:

$$s_{ax} = S/(2R) = 19.25/(2 \cdot 3.5) = 2.75\ mm$$

To obtain the equivalent depth, set the pad clearance ratio equal to:

$$\psi_v = \frac{(2.75 + 0.105)/2}{0.105} = 13.6$$

The curvature center should be middle sector angle to have the maximum bearing radial clearance in the center of the sector.

Define these parameters for each pad. To move between a pad and another use the arrows at the bottom.

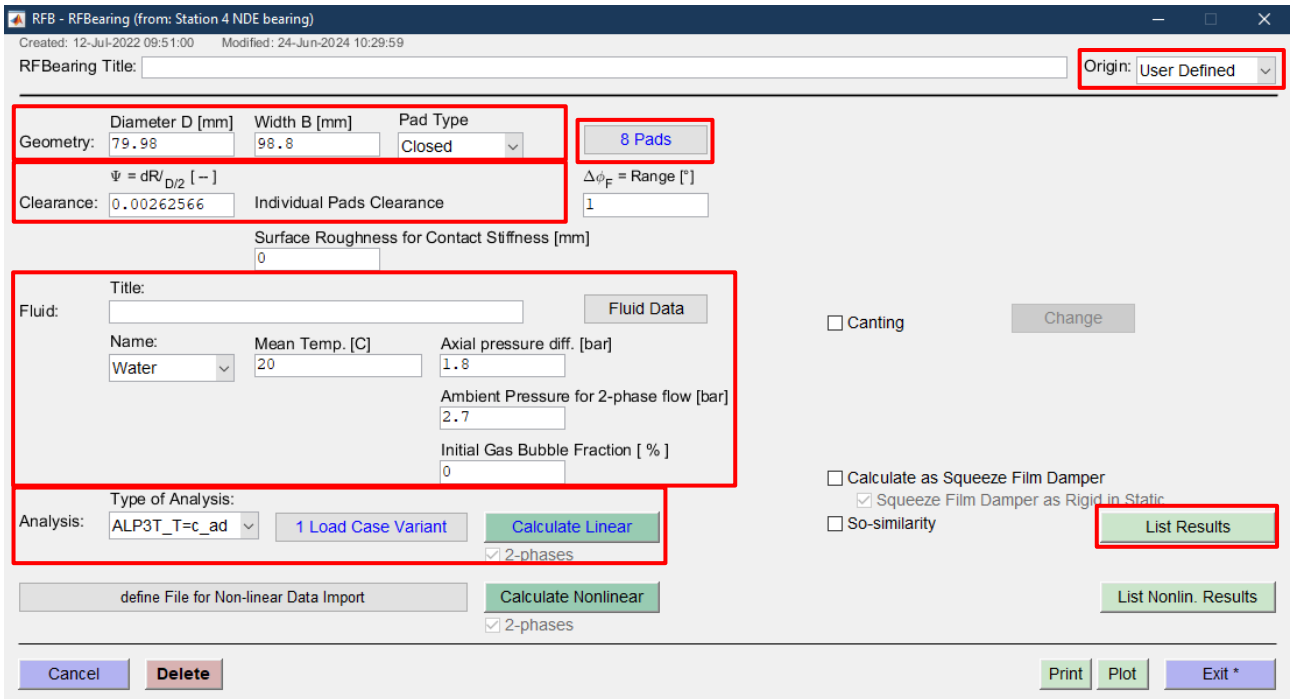

Figure 34.. RFB Object GUI: Product lubricated bearing.

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- Next RFB object requires fluid properties. PLB use as lubricant the operating fluid. In most cases the operating fluid is water and properties are already available. If the operating fluid is not present in the list provided, edit a custom fluid with the correct properties.
  Insert the absolute pressure at the end of the bearing and the pressure drop in it.
- In "Analysis" select as types of analysis: ALP3T_T=c_ad (constant adiabatic) when the lubricant is water. The fluid film temperature is constant. It is input as a mean temperature. This analysis also takes into account turbulence. The Reynolds number therefore is another dependency apart from the Sommerfeld number for this analysis type.
  This analysis considers 2-phase flow for the cavitation zones that is important when working with water and discharge pressure near ambient pressure.
- The results can be listed by the button "List Results".

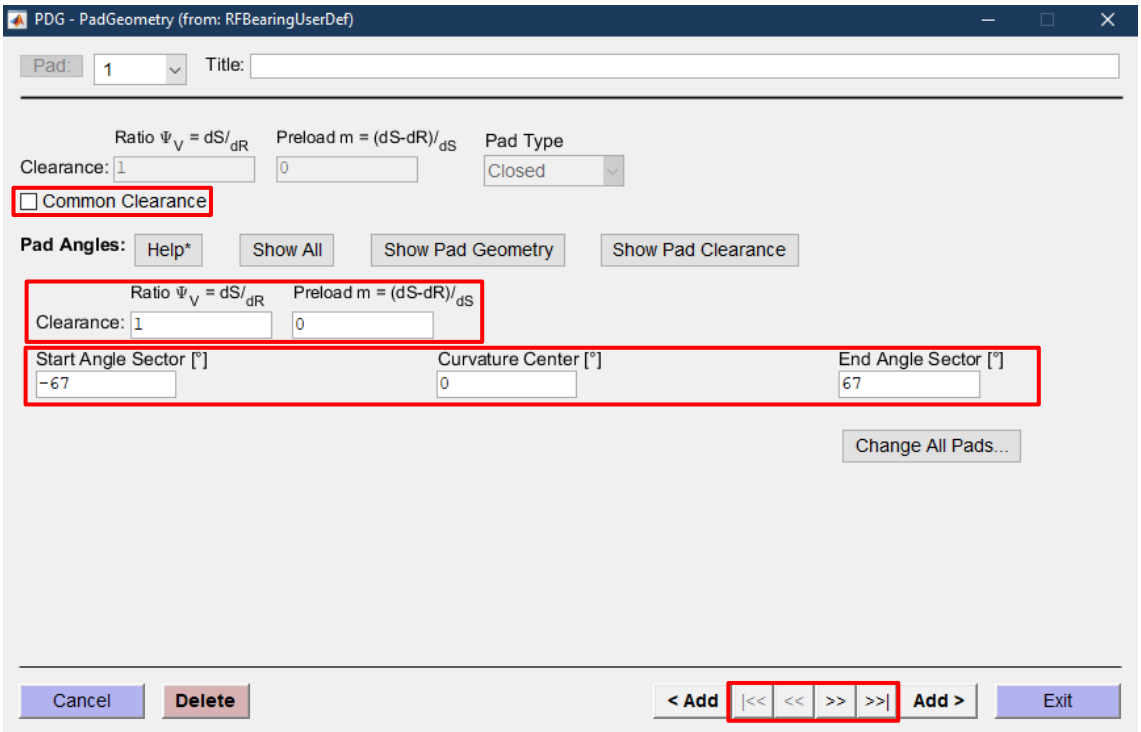

Figure 35. Pad Angles GUI: Product lubricated bearing, pad region.

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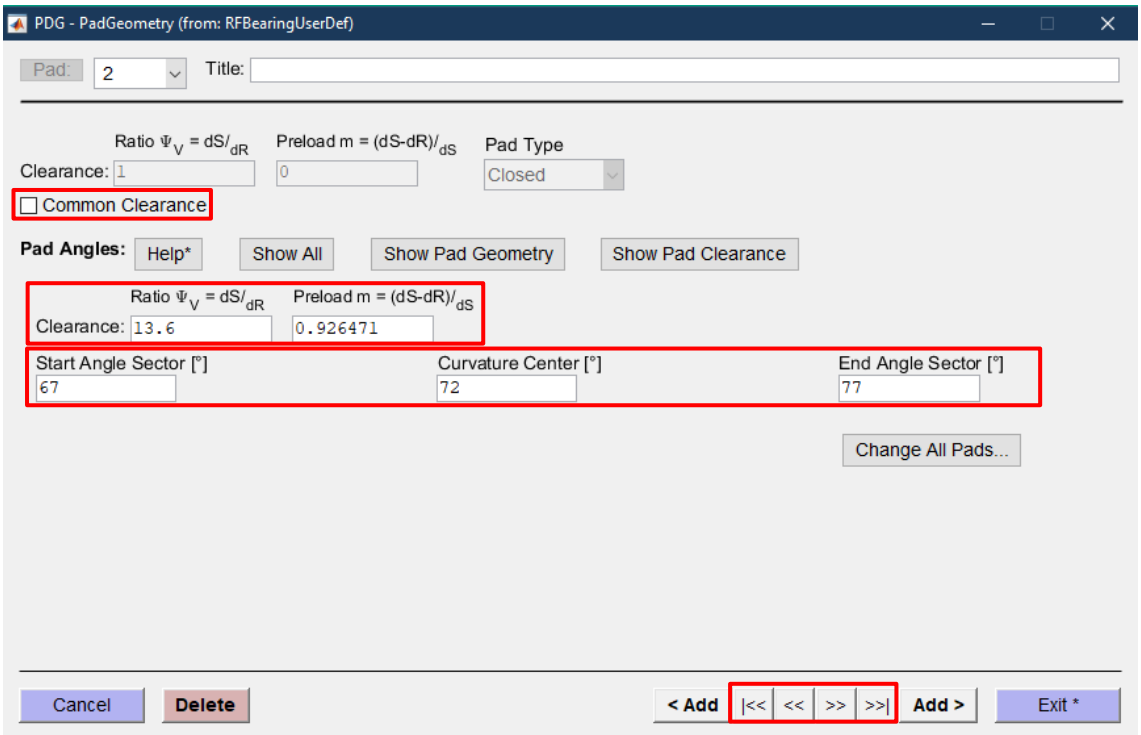

Figure 36. Pad Angles GUI: Product lubricated bearing, groove region.

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1.5.1. Madyn 2000, Hydrodynamic pockets

Internal made journal bearings (Figure 37) can have in the lower bushing a circumferential groove that can be approximated in Madyn 2000 as a hydrodynamic pocket.

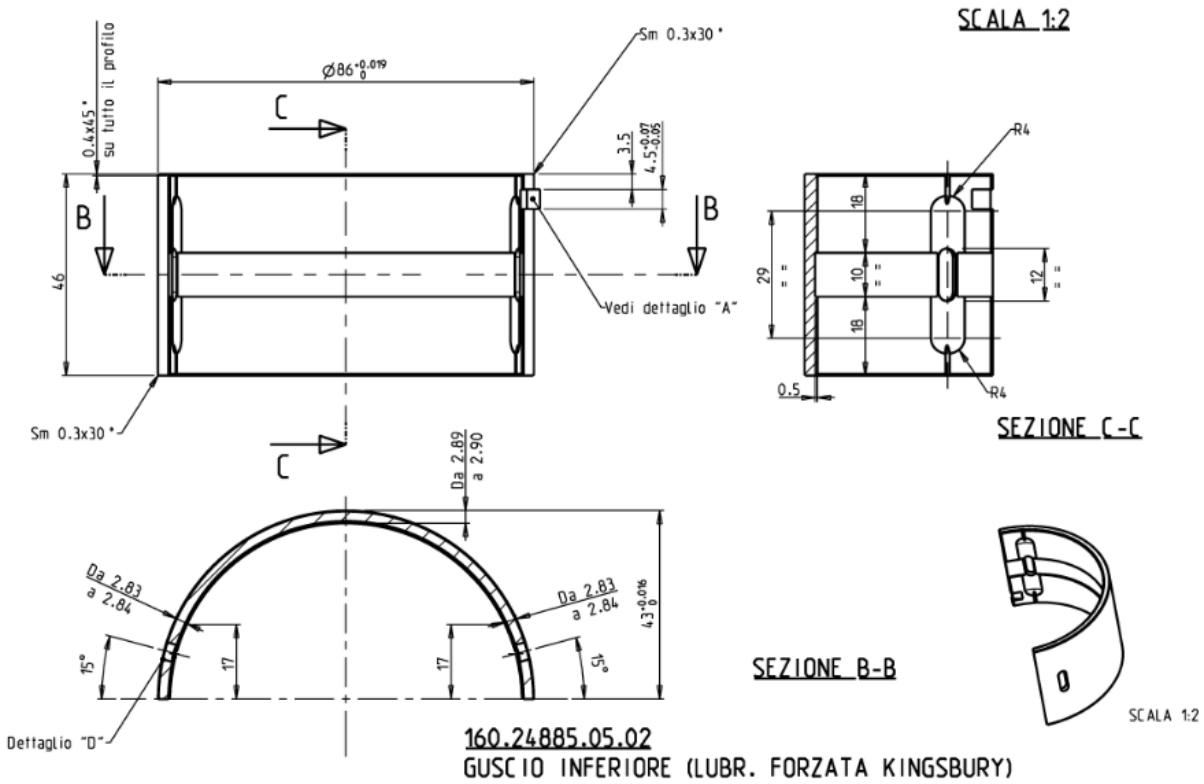

Figure 37. Internal made journal bearing.

In the Pad Angles GUI (Figure 38) set the "Hydrodynamic Model" as *Pockets*. The hydrodynamic pocket in the current pad will start at the *Start Angle at Pad End* of the previous pad and will terminate at the *End Angle at Pad Start*. Use the same Curvature Center of defined for the pad.

In Figure 37 width is equal to 10 mm and theorical depth to 0.5 mm. Pocket position 50% means, that it is axially centred. Note, that for constant groove depth the clearance ratio Ψv and theorical depth must be consistent. The clearance ratio for the groove is:

$$\psi_v = \frac{(th.depth + dR)}{dR} = \frac{(0.500 + 0.052)}{0.052} = 10.6$$

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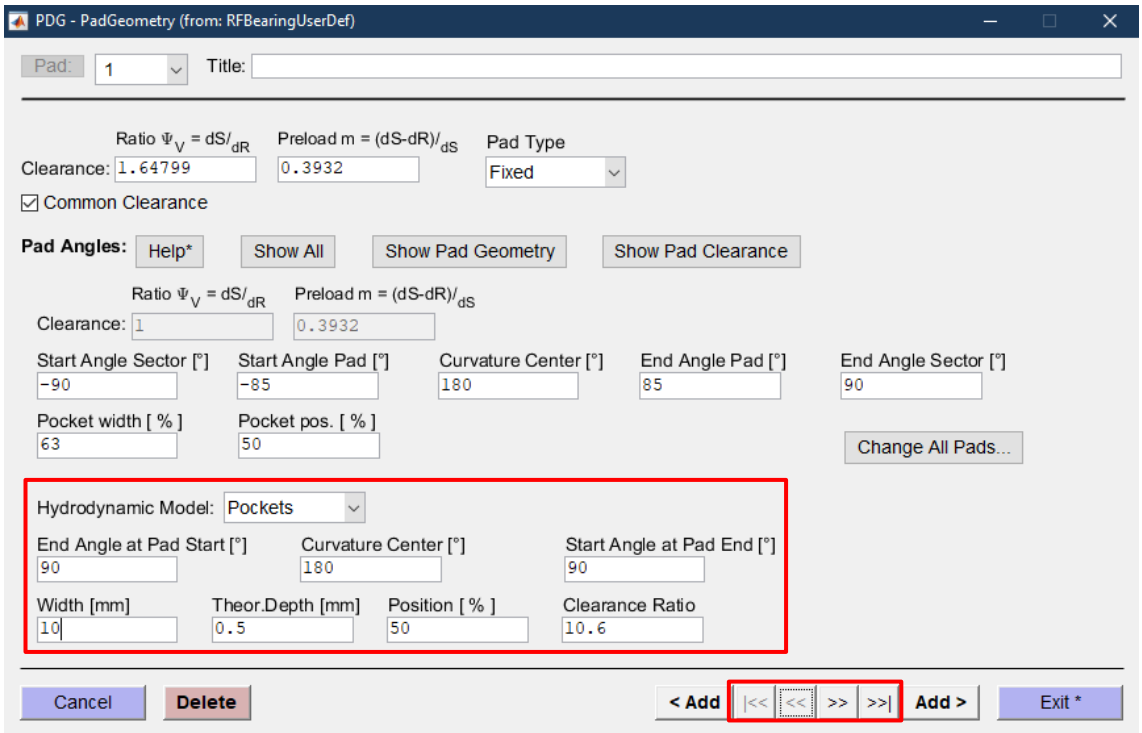

Figure 38. Pad Angles GUI: Hydrodynamic pocket.

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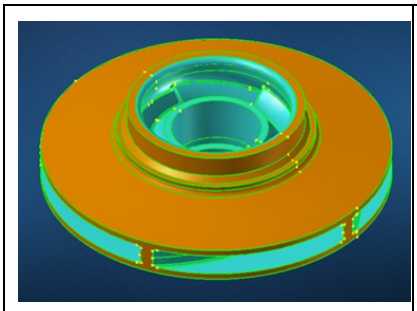
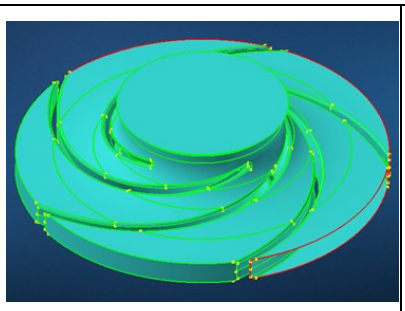
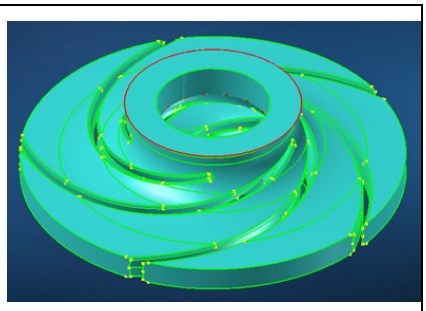
1.6. MSC Apex, Fluid added mass

An easy way to estimate the **fluid trapped in impeller passages** is to upload the 3D model, if available, in MSC Apex and create the fluid domain starting to the metal as follow:

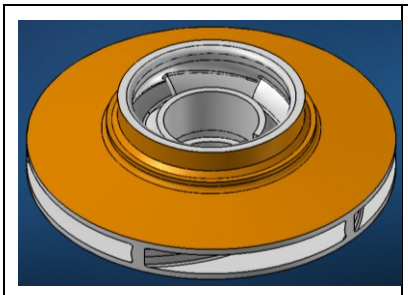
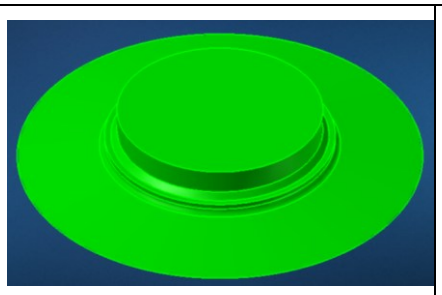
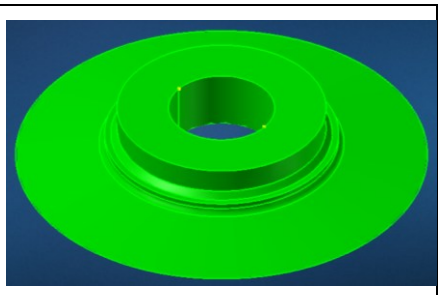
- select and delete all impeller external surfaces.
- solidify the new impeller surface by closing all openings with the command “Filler”.
- remove the regions that do not represent the fluid domain (for example the region occupied by the shaft) The commands “Split Surface” and “Push/Pull” can help in this operation.

		
Delete all external surfaces	Solidify the geometry	Remove the metal regions

In the “Query tools” the command “Mass Properties of Solids” can be used to estimate the volume of trapped fluid. It will be multiplied by the fluid density to obtain the trapped mass.

To estimate the fluid swept consider two contributions: fluid swept by impellers and immersed shafting. **Fluid swept by impellers** can be evaluated in a similar way as fluid trapped in impeller passages using MSC Apex:

- select and create impeller surfaces of front and back shroud with the command “MidSurface”.
- solidify the shrouds by closing all openings with the command “Filler”.
- remove the regions occupied by the shaft to not consider it twice. The commands “Split Surface” and “Push/Pull” can help in this operation.

		
Create surface of front and back shroud	Solidify the geometry	Remove the shaft region

Estimate the mass swept by impellers as explained for the trapped fluid.

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The **fluid swept by immersed shafting** can be directly included as mass per unit length in the shaft spreadsheet in the “addmass” column (1.2 Madyn 2000, Shaft object). It will be equal to:

$$m/L = \rho_f \frac{\pi}{4}(ODmb^2 - IDmb^2) \quad [kg/m]$$

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1.7. Madyn 2000, Static analysis SAN

The types of loads in a static analysis (Figure 39) are:

- **Gravitation.** The magnitude and direction of acceleration of gravity has to be defined. Default is 9.81 m/s^2 in 2-direction. The mass of spring bearing supports is not considered in the analysis (it is considered a dynamic mass).
- **Static load case.** Two types of loads are mainly used:
    - **Forces and Moments (**Figure 40**).** It allows to apply force and moments at selected stations. The load can be speed dependent.
    - **Radial Bearing Support Displacement (**Figure 41**).** The displacement of the support is prescribed. It can be used when applying a lift to the rotor.
- **Static load case combination.** It allows to combine all types of available load case.

The static analysis has two main purposes (Figure 42):

- Evaluate the static deflection with a “static deflection analysis”. In this case check the boxes Rigid Support that will consider the bearings infinitely stiff and Fluid Deactivated that means an analysis without fluid influence will be carried out (effect of wear rings or impeller interaction). The analyses with rigid support are speed independent.
  Select as load case, one in which the default gravitation load was defined.
  NOTE. To obtain a proper estimation of the static deflection do not include in the rotor model the fluid added mass.
- Estimate a reaction force acting on the bearings in operating condition with a “dynamic deflection analysis”. It is an iterative procedure in which annular seals are modelled as linear spring elements and bearing stiffness is iterated until bearing load and deflections reach convergence. The bearing load obtained will be used to estimate the dynamic coefficients in the dynamic analyses.
  Uncheck the boxes Rigid Support and Fluid Deactivated. The analysis will be speed dependent since the fluid film bearing properties and rolling element properties depend on the speed.
  Select the correct combination of loads for the analysis. For pumps in volute the load acting on bearings are gravity and radial thrust; while in diffusers the load acting is only related to gravitation and radial thrust can be neglected.

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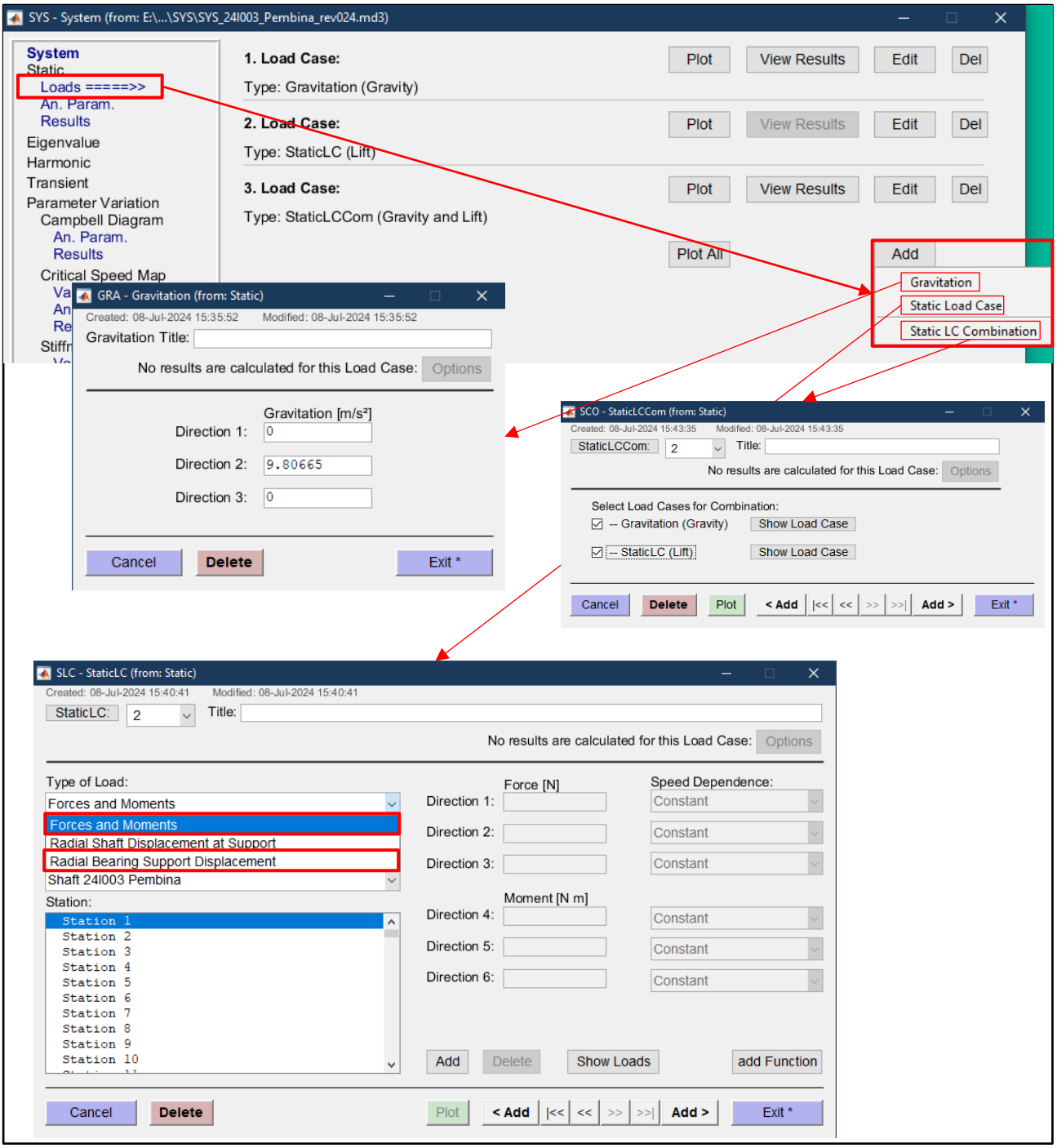

Figure 39. Static Analysis Type of Loads GUI.

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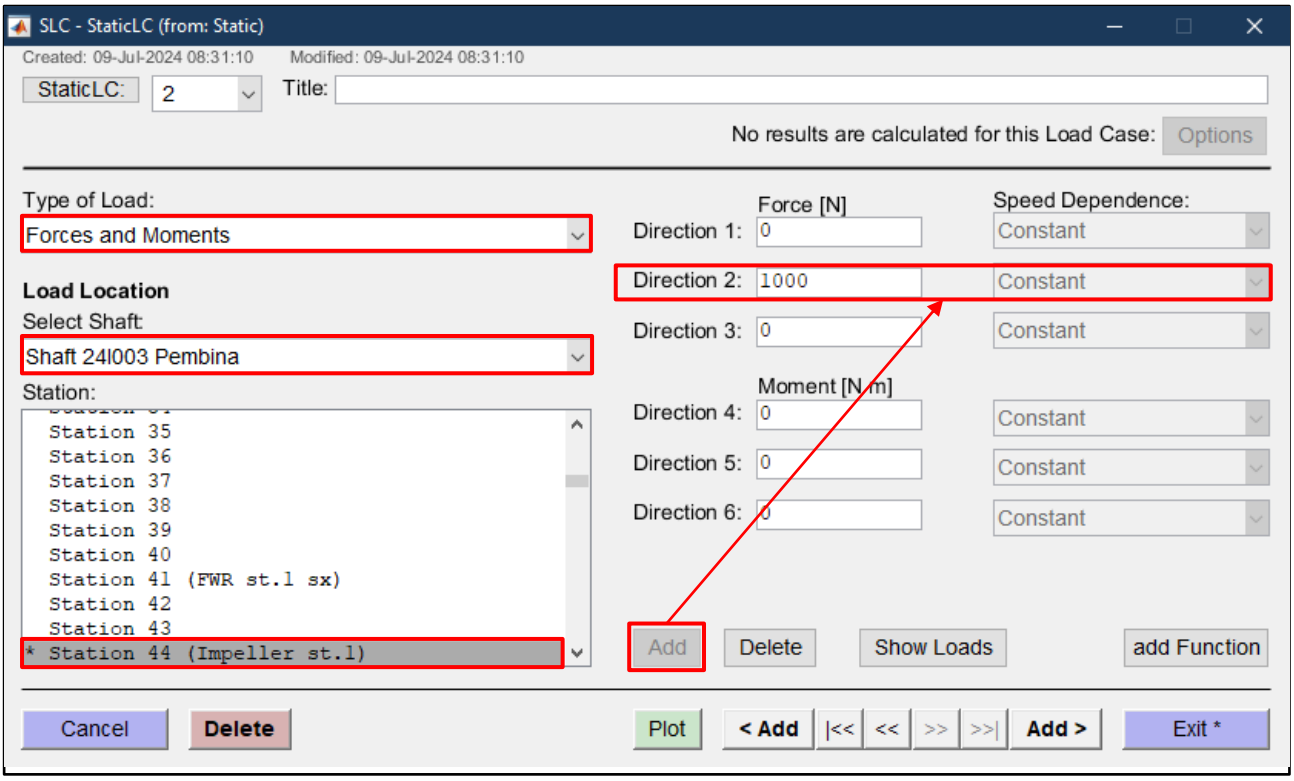

Figure 40. Static Load case, Forces and Moments GUI.

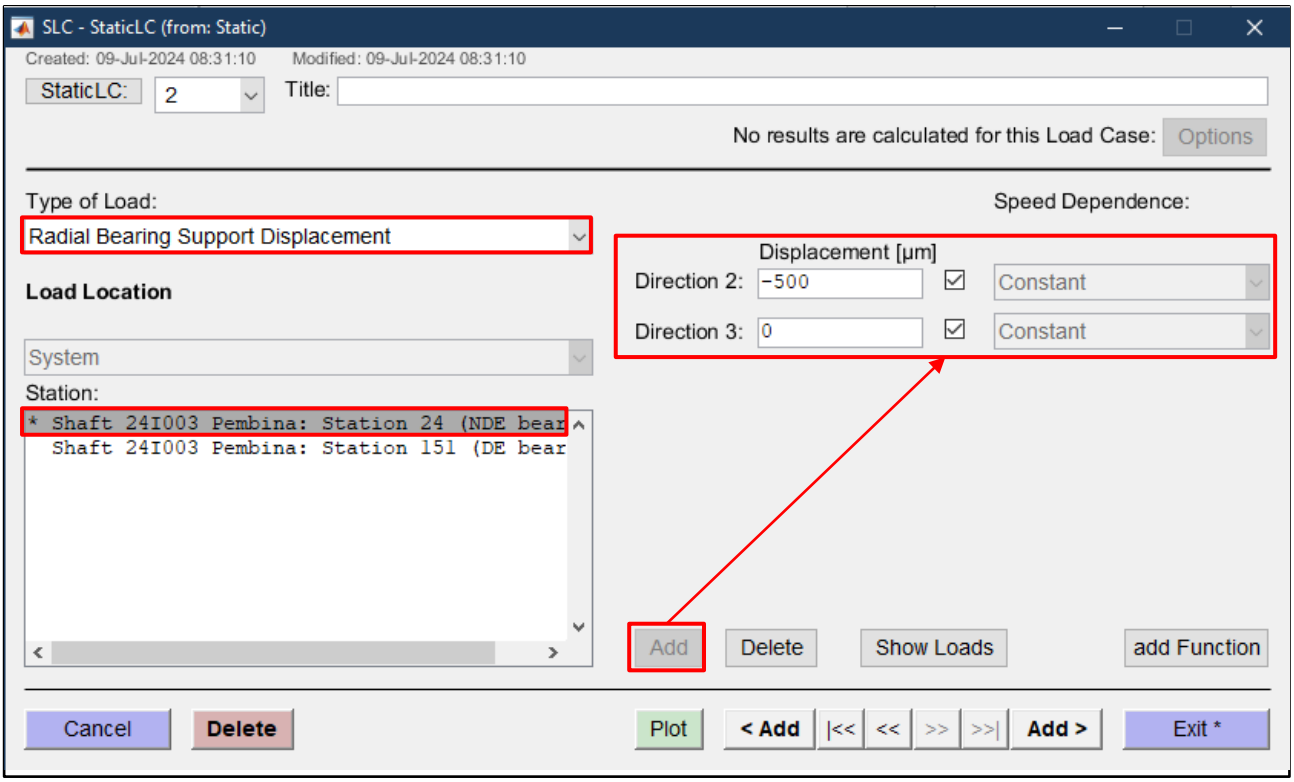

Figure 41. Static Load case, Radial Bearing Support Displacement GUI.

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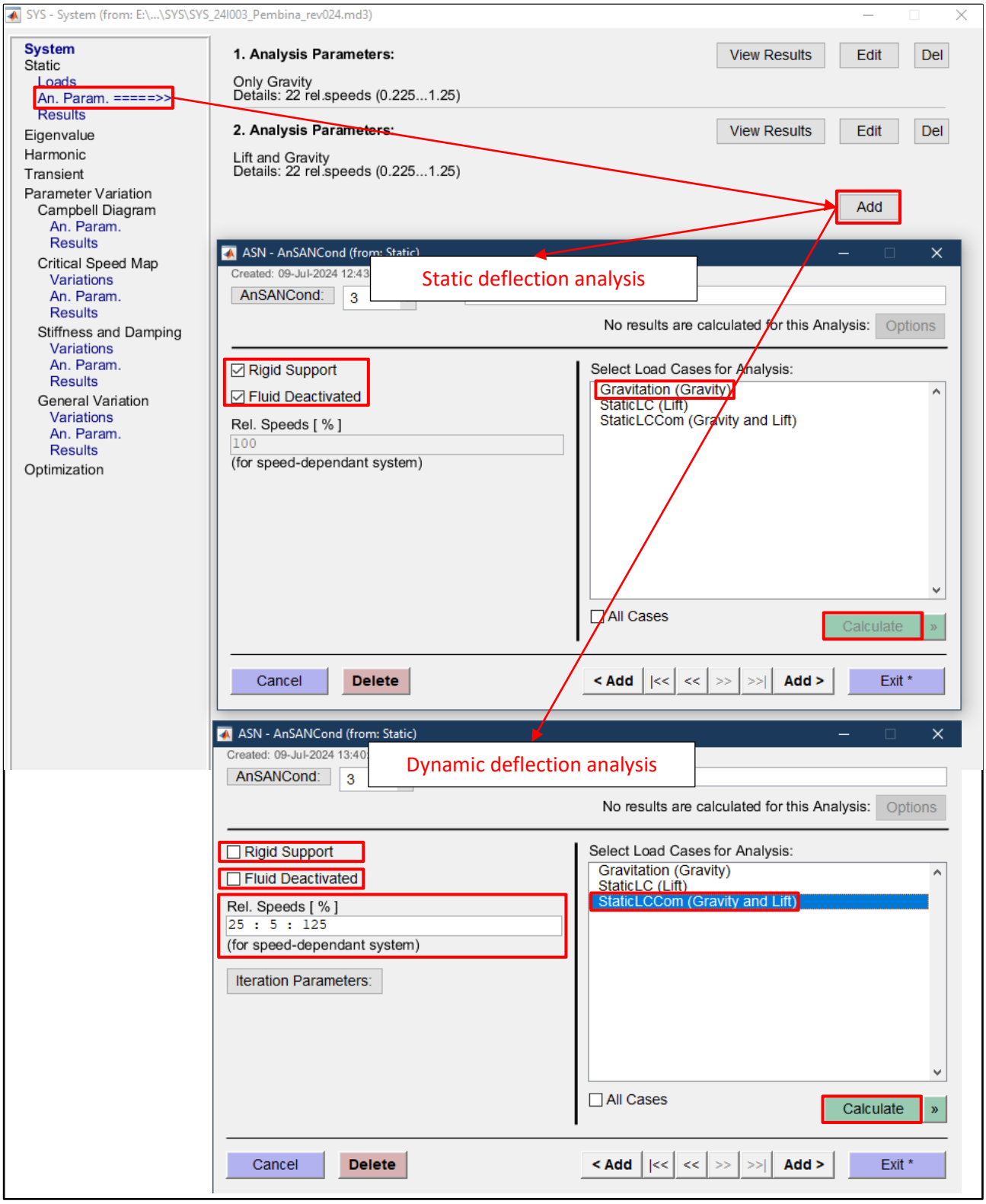

Figure 42. Static Analysis Parameters GUI.

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After performing static analysis, access the results as shown in Figure 43. Select the proper load case and plot the shape plot of selected result. In all shape plots the contour of the shaft is drawn in the background.

In static analysis results plotted are:

- Displacements;
- Forces and Moments;
- Stresses (components);
- Equivalent stresses.

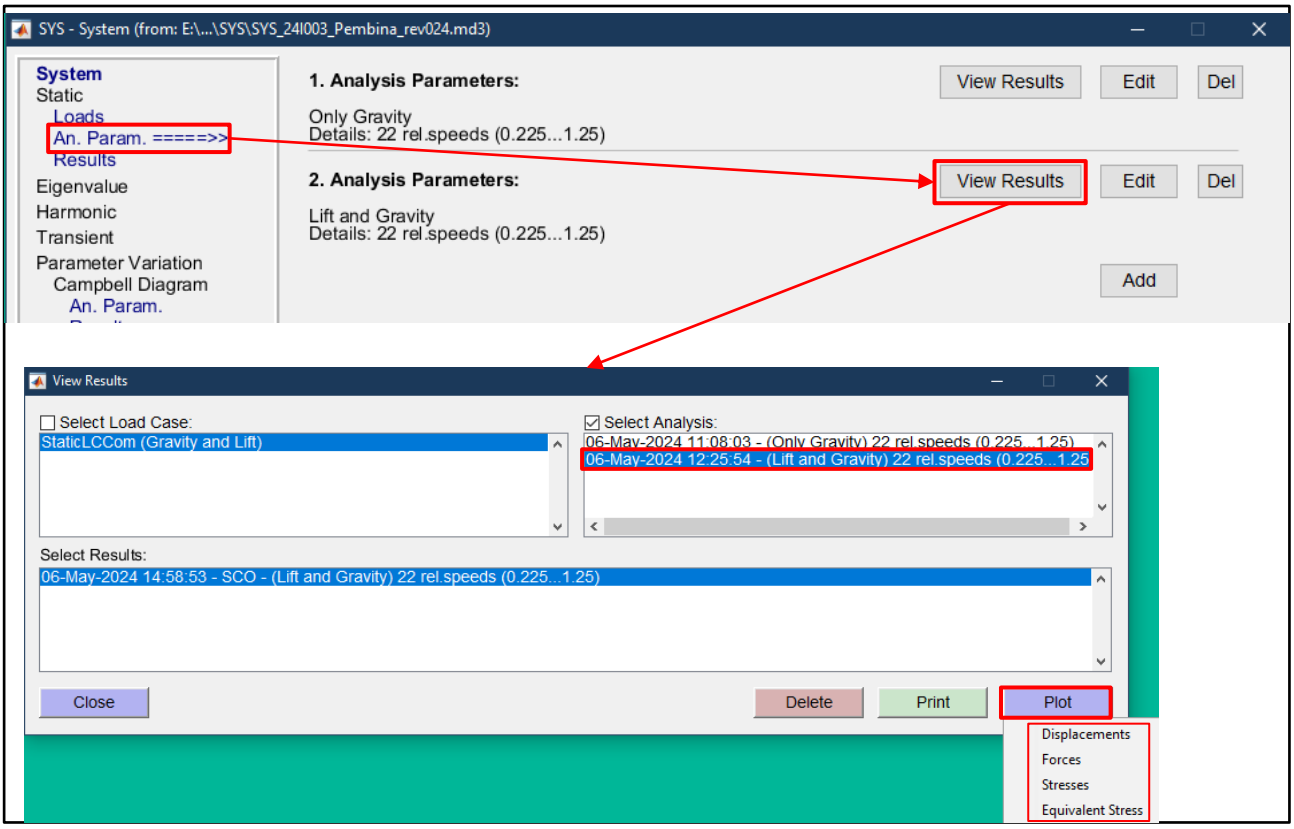

Figure 43. Static Results GUI.

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1.8. Madyn 2000, Parameter variation, Campbell diagram CDG.

Campbell diagram CDG is an eigenvalue analysis with the variation of rotor speed.

Parameters can be set for Campbell diagram analysis (Figure 44):

- In case RFBs or REBs are present, they can either be ignored or considered with a static load. The static load can be defined via the selection of static analysis result (suggested) or in case of RFBs with no REB in the system by direct load input. If RFB/FRB/REB loads are defined from results of a normal static analysis, which is speed dependent, it is necessary that all speeds are covered by the static analysis. Individual RFB/REB bearings can be ignored by individually disabling them.
- The maximum and minimum frequency for modes to be stored. Maximum frequency should be set greater than 2.2 times the maximum continuous speed
- The rotor speeds to perform the analysis. The speed range shall be calculated from 25% to 125% of rated speed. The notation $25{:}5{:}125$ will perform the analysis from 25% to 125% of rated speed with increment of speed of 5%.
- A damping ratio limit “Dlimit” for modes to be stored (suggested $Dlimit = 0.9$). For $Dlimit = 1$ all modes including real modes are stored, for $Dlimit < 1$ only modes with a lower damping ratio than Dlimit. Note, that for $Dlimit < 1$ real modes are deleted, also unstable real modes.
- A check box allows to eliminate the left eigenvectors for storage (suggested uncheck).
- A check box allows the possibility to deactivate fluids (suggested uncheck).
- A check box allows to select critical speed analysis. In this case the intersections of speed line with the mode frequency lines are calculated and the modes shapes are stored at these frequencies (suggested check).

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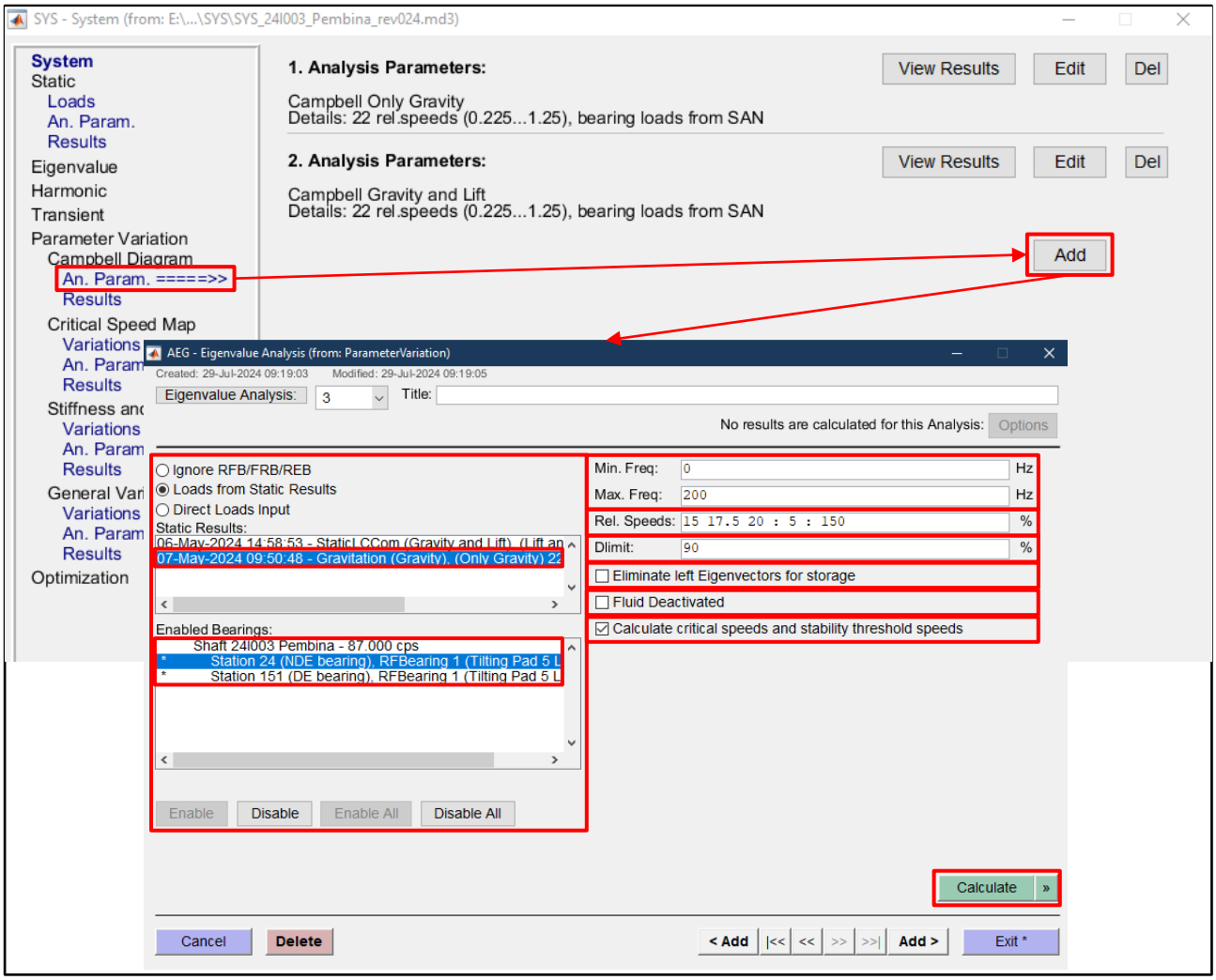

Figure 44. Campbell diagram GUI.

After performing the analysis, access the results as shown in Figure 45. Select the proper analysis and results and plot the diagram and shape plot of selected result.

In “Diagram Plot” a Campbell diagram with two subplot is shown. The top plot shows the natural frequency trend of the estimated modes in function of the rotating speed. The bottom plot shows the damping ratio trend of the estimated modes.

The “Shape Plots” show the mode shapes of the different modes at different rotating speeds.

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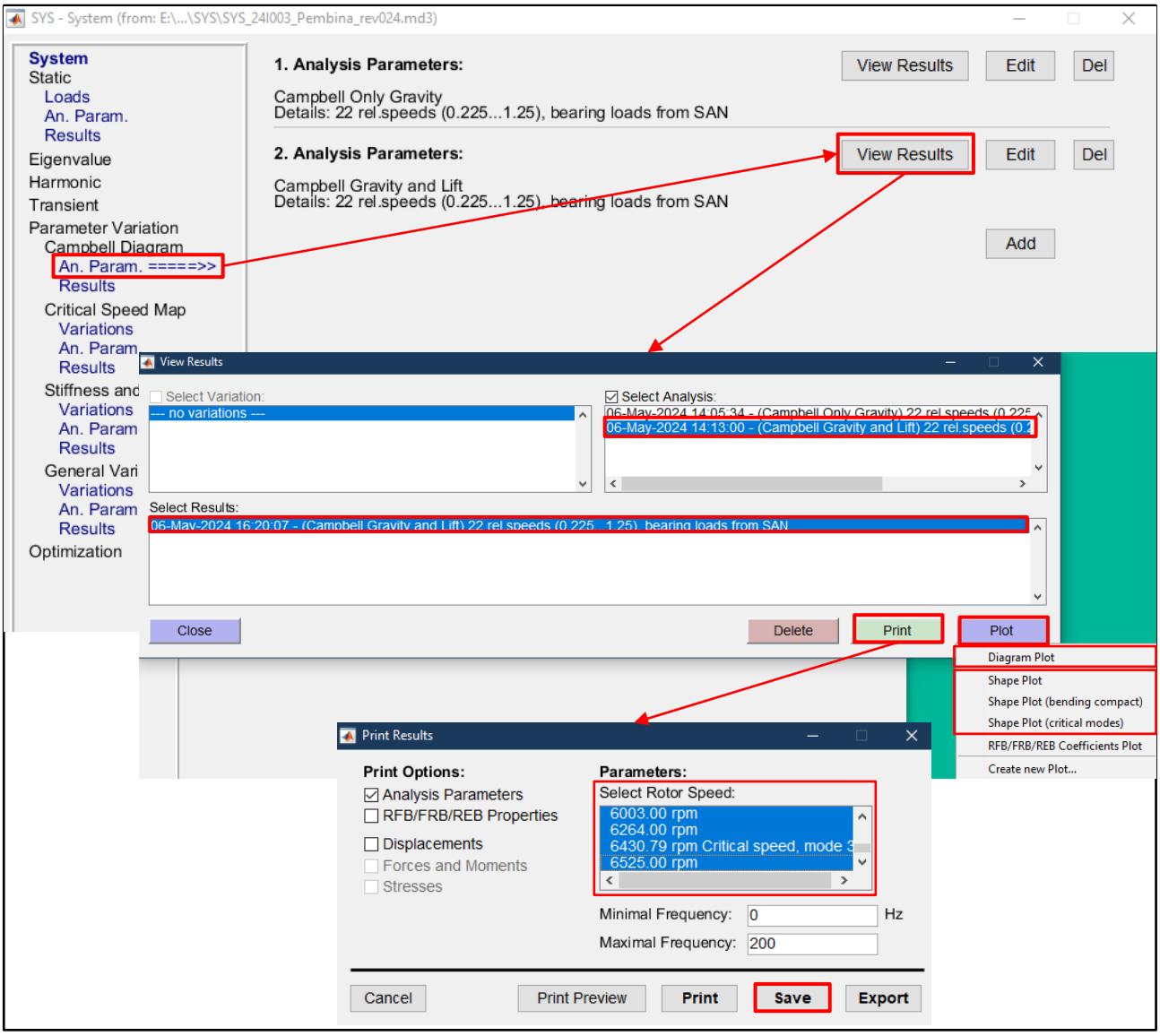

Figure 45. Campbell diagram Results GUI.

As Annex I API 610 12th Edition recommend, for both as-new and 2x as-new clearances, the damping factor versus the separation margin between any bending natural frequency and the synchronous run line shall be within the “acceptable” region (Figure 46). Refer to Mod.XXX-Annex I Diagram to verify if this condition can be satisfied. If not, the damped response analysis shall be determined.

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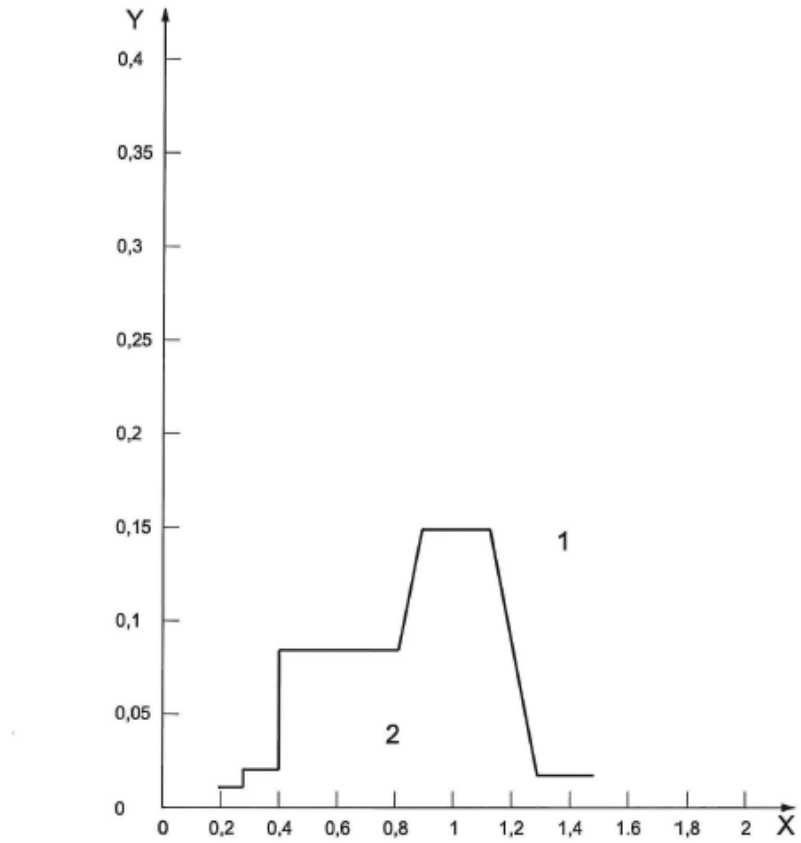

Key
X frequency ratio, f_{nf}/f_{run}
Y damping factor, ξ
1 acceptable region
2 unacceptable region

Figure 46. Damping Factor vs Frequency Ratio

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Annex C. Report structure.

Report should include:

1. INTRODUCTION.

- a brief description of the machine in study (type, train components and rotating speed);
- the Standard reference (i.e. API 610 12th Edition, January 2021) and criteria to satisfy (API 610 §9.2.4.1.2 and Annex I).

2. NUMERICAL MODEL.

It should provide the assumption made to model system, a schematic diagram of the model depicting nodes and elements and the acceptance procedure for the lateral analysis described API 610 Annex I.

3. RESULTS.

3.1. DRY CONDITION. A Campbell diagram should show the rotor's first, second and third "dry" bending natural frequency.

Figures of the bending mode shapes should be included.

(If the first bending natural frequency has a separation margin higher 20% of the highest excitation frequency no other analysis are required).

3.2. AS NEW CLEARANCE.

3.2.1. WITH WATER. A Campbell diagram should show all the rotor's damped natural frequencies with water as operating fluid within the frequency range zero to 2.2 times maximum continuous speed (25% to 125% of rated speed).

Figures of the bending mode shapes should be included.

Annex I diagram should report the damping factor versus separation margin between bending natural frequency and the run line.

3.2.2. WITH OPERATING FLUID. A Campbell diagram should show all the rotor's damped natural frequencies with the operating fluid within the frequency range zero to 2.2 times maximum continuous speed (25% to 125% of rated speed).

Figures of the bending mode shapes should be included.

Annex I diagram should report the damping factor versus separation margin between bending natural frequency and the run line.

3.3. TWICE THE AS NEW CLEARANCE. A Campbell diagram should show all the rotor's damped natural frequencies with the operating fluid within the frequency range zero to 2.2 times maximum continuous speed (25% to 125% of rated speed).

Figures of the bending mode shapes should be included.

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Annex I diagram should report the damping factor versus separation margin between bending natural frequency and the run line.

(If the damping ratio of any natural frequency is in the acceptable region of Annex I diagram no other analysis are required).

3.4. RESPONSE TO UNBALANCE. Figures should show the unbalance distribution for the modes shapes inside the not acceptable region of API Annex I. A table should state the peak-to-peak displacement of the unbalanced rotor at the points of maximum displacement should not exceed 35% of the diametral running clearance at that point.

4. CONCLUSIONS. It should contain a summary of all analysis done and state if the system meets the requirements.

5. ANNEX. It is a compendium of the numeric model.

5.1. INERTIA TABLE that defines the position and the inertia properties of the lumped masses.

5.2. MATERIAL PROPERTIES. A table that includes density, elastic modulus and Poisson's ratio of all materials used in the analysis.

5.3. ROTOR EQUIVALENT RIGID BODY PROPERTIES. A table that shows the main properties of each rotor (total length, mass, moment of polar inertia, speed ratio).

5.4. SHAFT ELEMENTS. Tables that define the numerical model geometry (lengths, outer diameters, inner diameters of each element) of the shafts in the system.

5.5. BEARING COEFFICIENTS. A table containing the characteristics of bearing (type, lubricant, diameter, …). The position of the bearings and the dynamic coefficient of all bearings at different rotating speeds.

5.6. LIQUID ANNULAR SEALS. The position of each seal should be reported. A table containing the dynamic coefficients of each seal at different rotating speed should be included for the different conditions analyzed: as new clearance with water, as new clearance with operating fluid and twice the as new clearance with operating fluid.

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